PRODUCER GUIDE

Ratio Vitis 0.1.0 · Ratio Complexus

The vineyard *you measure* from soil to air

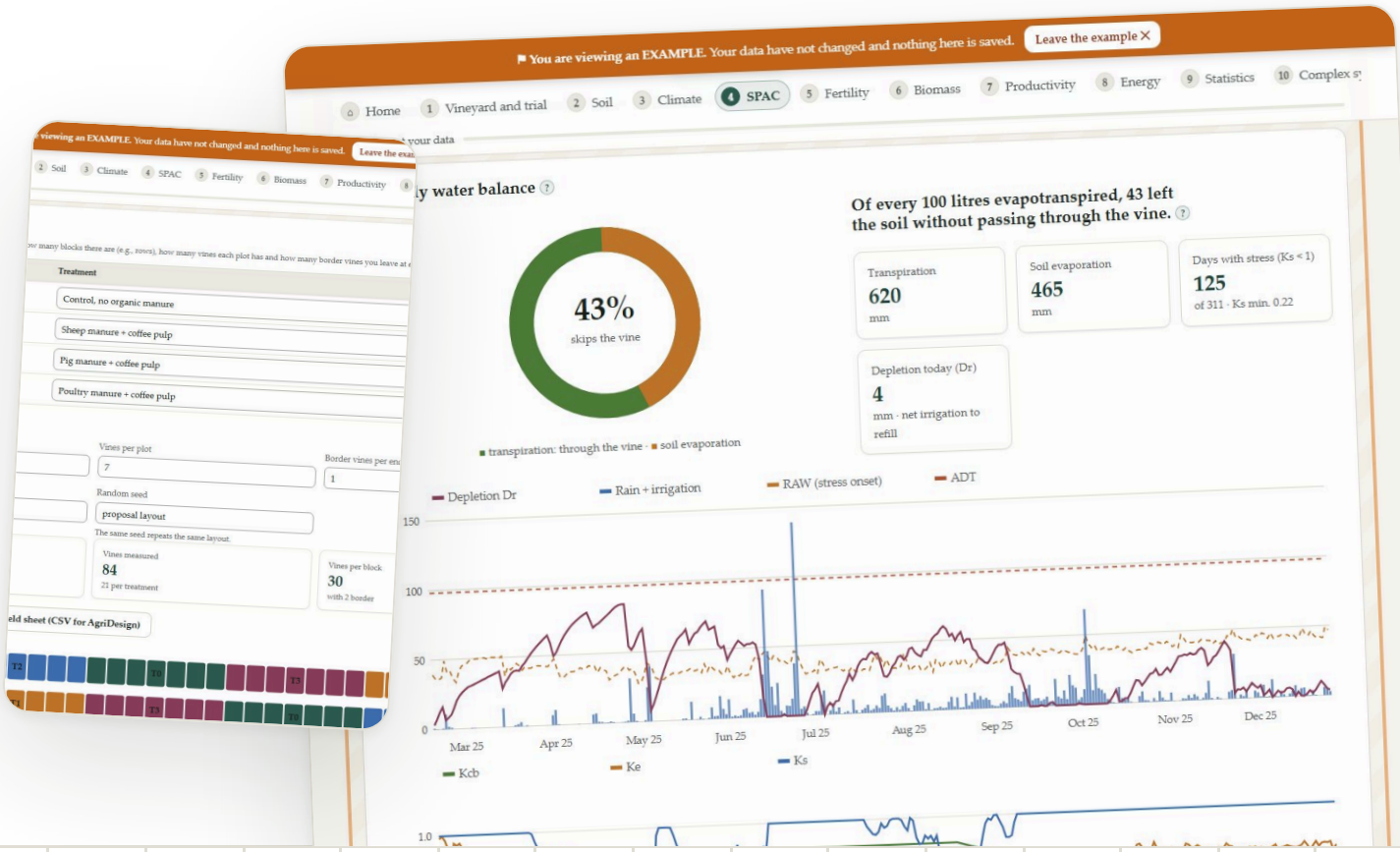
A practical guide to measuring your vineyard with Ratio Vitis —soil, climate, water, nutrition, biomass, harvest and energy— and reading it as a system you can compare and project.

1st edition · November 2026 · covers Ratio Vitis 0.1.0

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Héctor Tecumshé Mojica-Zárate · Jorge Hernández-González · Oscar Díaz-José · Adán Guillermo Ramírez-García

There is a companion guide for research and teaching: the Scientist Guide.



1 Vineyard and trial
2 Soil
3 Climate
4 SPAC
5 Fertility
6 Biomass
7 Productivity
8 Energy
9 Statistics
10 Complex system
11 Dashboard
12 Regeneration
13 Risks
14 Sustainability
15 Outlook
16 Report

BEFORE READING

How to use this guide

Who it is for, how it is organised and what each sign means.

This guide is for whoever works a vineyard and wants to measure it: growers, field technicians, students doing their thesis on a cooperating plot. You need no coding or statistics; you need a tape measure, a jar, a scale and the will to take notes.

Ratio Vitis is the quantitative sister of Ratio Viña. Ratio Viña gathers what people know about the vineyard; Ratio Vitis measures what happens in it: the water coming in as rain and irrigation, the water the soil holds, the water rising through the vine and the water the air takes away; the sun's energy, the carbon kept in the plant and the grapes that leave. Every number comes from your field or from a cited source.

It follows the app's order: one chapter per block, 1 to 16, each with its colour. Every example uses **one real case**: the cooperating vineyard of **Mata de Zarza, Puente Nacional, Veracruz**, with Perlet table grapes planted in February 2025 and the manure trial of Jorge Hernández-González's thesis.

Conventions

WRITTEN AS	MEANS
Draw again	An app button or field, with its text as it appears.
Report › Save project	A path: open the first item, then the next.
1.40	A value you type as shown.
1	A marker on a screenshot; its explanation is below with the same number.
● Published data · ◆ Ratio Vitis reading · no data	The app's own labels: sourced data, flagged editable assumption, or something not yet measured.

The five boxes

◆ **IN MATA DE ZARZA**

The worked case, with the real numbers the app gives.

✓ **TIP**

Field shortcuts and good practice.

! **CAUTION**

Common mistakes that lead to a wrong conclusion.

? **WHY?**

The reason behind a recommendation. You can skip it.

→ **DECISION RULE**

If you see this, do that.

? **RATIO VITIS DOES NOT MAKE UP DATA**

Whatever has no measurement or source shows as **no data**. In the Mata de Zarza case the manure trial has not been measured yet: that is why the soil, fertility, biomass, harvest and sustainability blocks look empty in the screenshots. This guide tells you what to enter in each.

GETTING STARTED

0 Open the app and meet the screen

Ratio Vitis is a web page that runs on its own in your browser. Nothing to install, no account, and your data goes nowhere.

0.1 Opening Ratio Vitis

Go to **hectortecumshe-ai.github.io/RatioVitis** from a computer, tablet or phone, or scan the QR code on the "Ratio Vitis on one page" page. It is a single file: it works offline while the tab stays open, needs no installation and no account.

? **WHERE IS MY DATA KEPT?**

Only in the browser you use; nothing is uploaded. To move it to another computer or to your adviser use Report › Save project and on the other one, Report › Open project.

0.2 The home screen

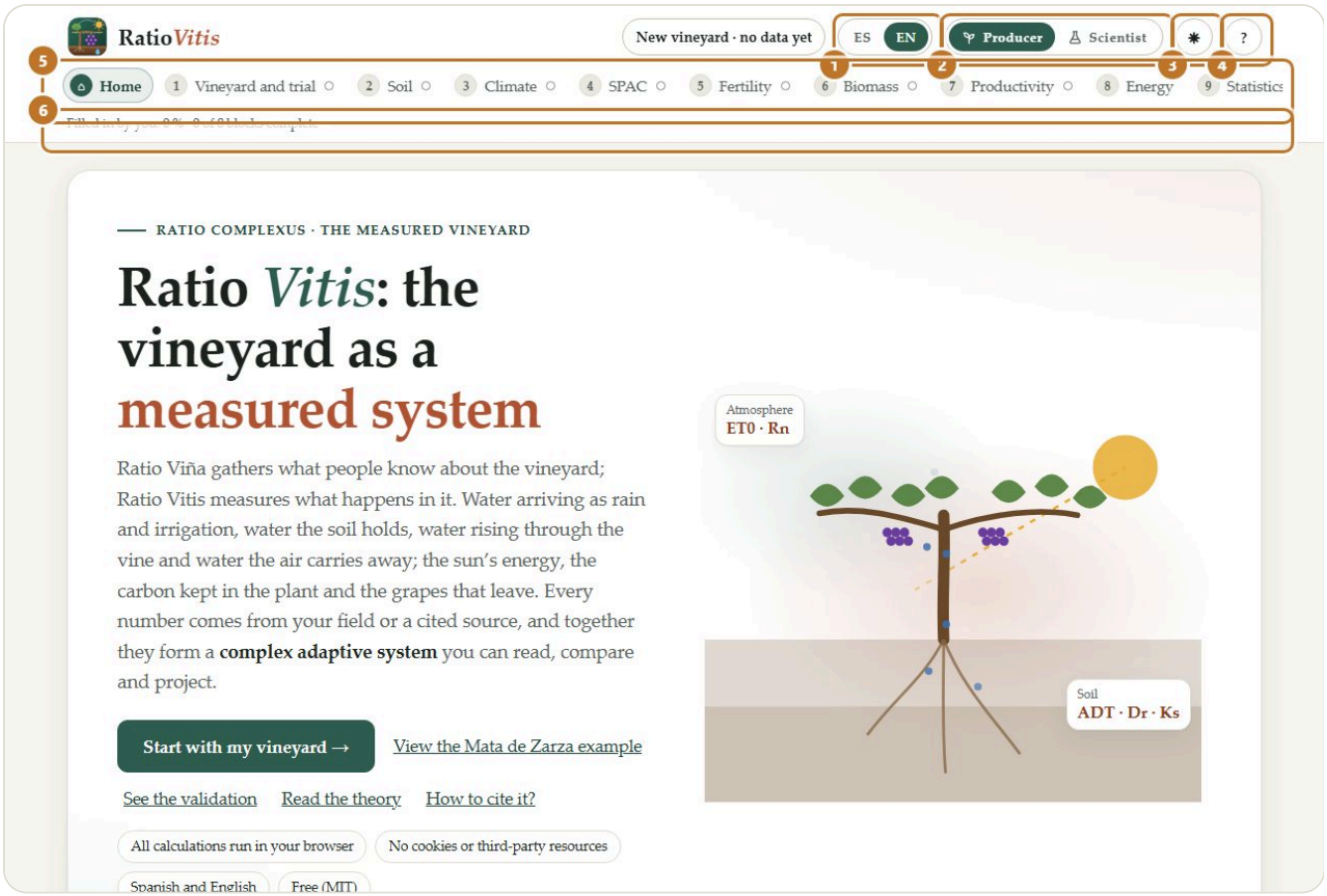


FIGURE 0.1 **Home.** Everything you need to get around is at the top.

1. **ES / EN** switches the whole app's language, report included.
2. **Producer / Scientist** switches the view; your data is kept.
3. ☀ / ☾ day or night theme.
4. **?** opens the general help.
5. The **block bar**, 1 to 16, plus Validation and Theory. Next to each block you fill, a mark says whether it is pending, in progress or complete.
6. **Filled in by you**: counts only what you entered, never the example or suggestions.

0.3 The "?" helps

Next to every important figure there is a **?**. Hover or tap it to see what it means, how it is computed and how to read it, with its source.

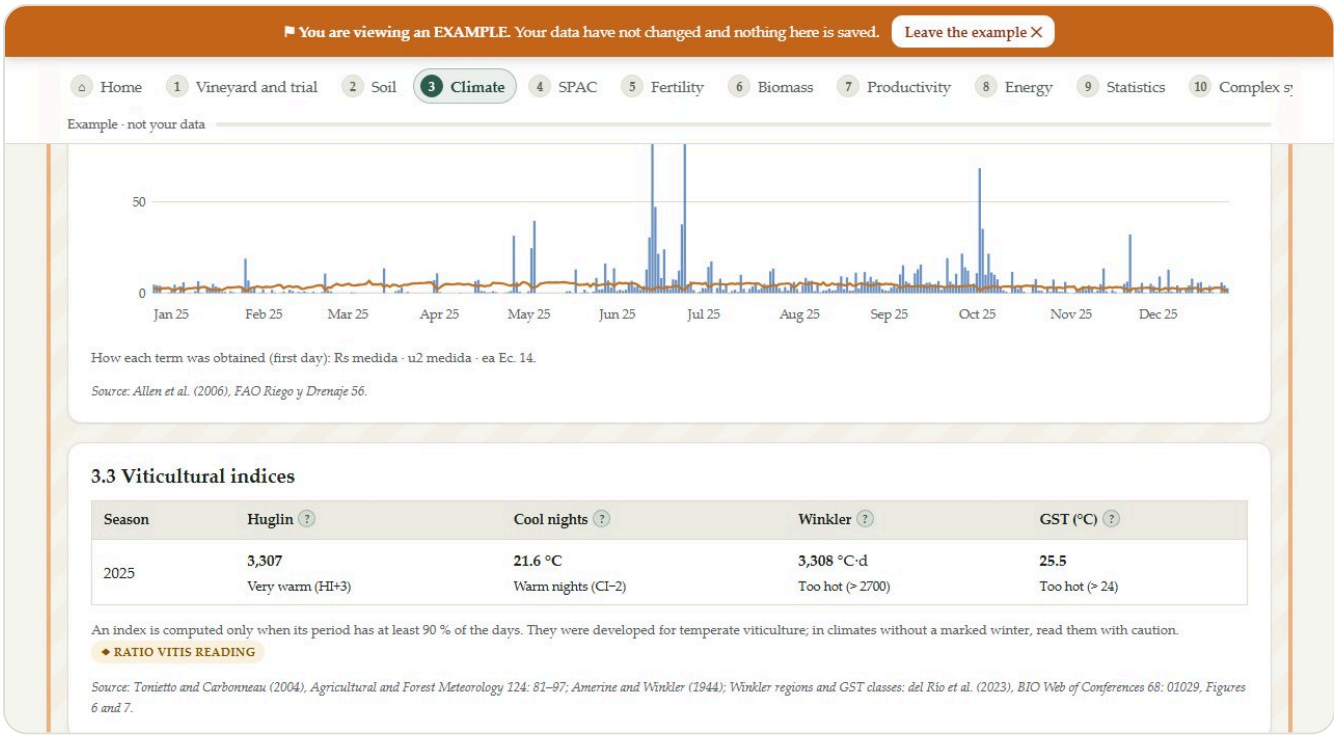


FIGURE 0.2 **Help for a figure.** The Winkler "?" in Block 3, with regions I to V.

0.4 Producer and scientist views

The **producer view** shows the essentials in plain language, with "In a few words" and "What to do" at the top of each block; it is the one this guide describes. The **scientist view** adds equations, assumptions, full tables and exports; the Scientist Guide explains it.

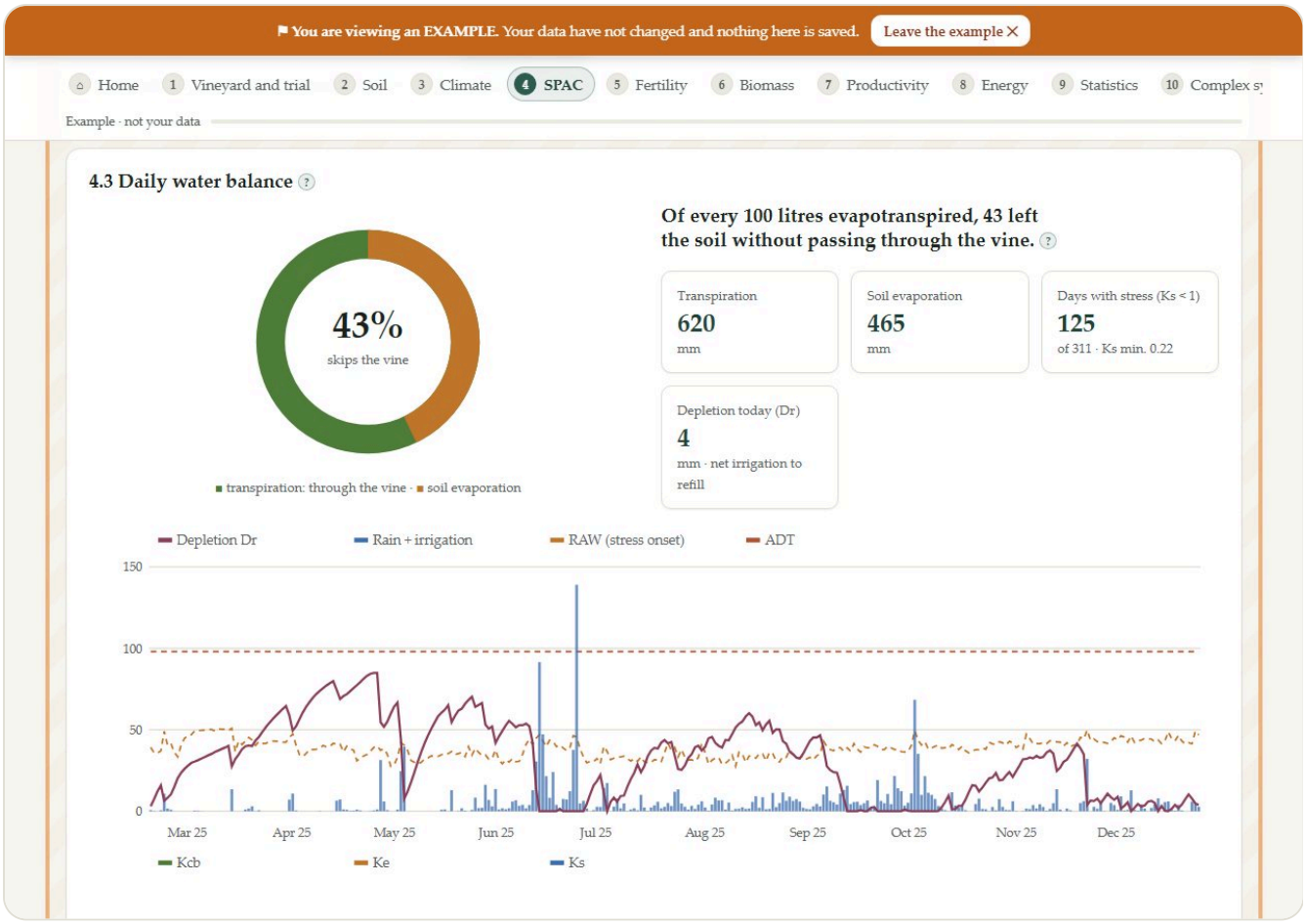


FIGURE 0.3 **Scientist view (in English).** The Block 4 water balance with its equations.

0.5 The full route

BLOCK 1 — 10 min
Vineyard and trial
Site, spacing and draw

BLOCK 2 — 10 min
Soil
The water tank

BLOCK 3 — 5 min
Climate
What the air demands

BLOCK 4 — 10 min
SPAC
Water from soil to air

BLOCK 5 — 8 min
Fertility
Leaf, soil and manures

BLOCK 6 — 8 min
Biomass
Vigour, canopy and carbon

BLOCK 7 — 6 min
Productivity
Fruit, quality and water

BLOCK 8 — 3 min
Energy
How sunlight is shared

BLOCK 9 — 10 min
Statistics
Real difference or chance?

BLOCK 10 — 5 min
Complex system
The vineyard network

BLOCK 11 — 3 min
Dashboard
All on one page

BLOCK 12 — 6 min
Regeneration
Practices and outcomes

BLOCK 13 — 6 min
Risks
Likelihood × impact

BLOCK 14 — 8 min
Sustainability
Environmental, economic, social

BLOCK 15 — 6 min
Outlook
At 1, 2 and 5 years

BLOCK 16 — 3 min
Report
Actions, PDF and data

Blocks 1 to 9 are for **entry**: that is where you type what you measure. Blocks 10 to 16 are **results**: they gather what you entered and read it as a system. You can jump between blocks at any time; whatever is missing shows as “no data”.

0.6 The Mata de Zarza example mode

The app starts blank: it never comes pre-filled. To first see a real vineyard, tap **View the Mata de Zarza example** on Home. You enter **example mode**: an orange band reminds you those data are not yours and nothing is saved; **Leave the example** takes you back to your project as you left it.

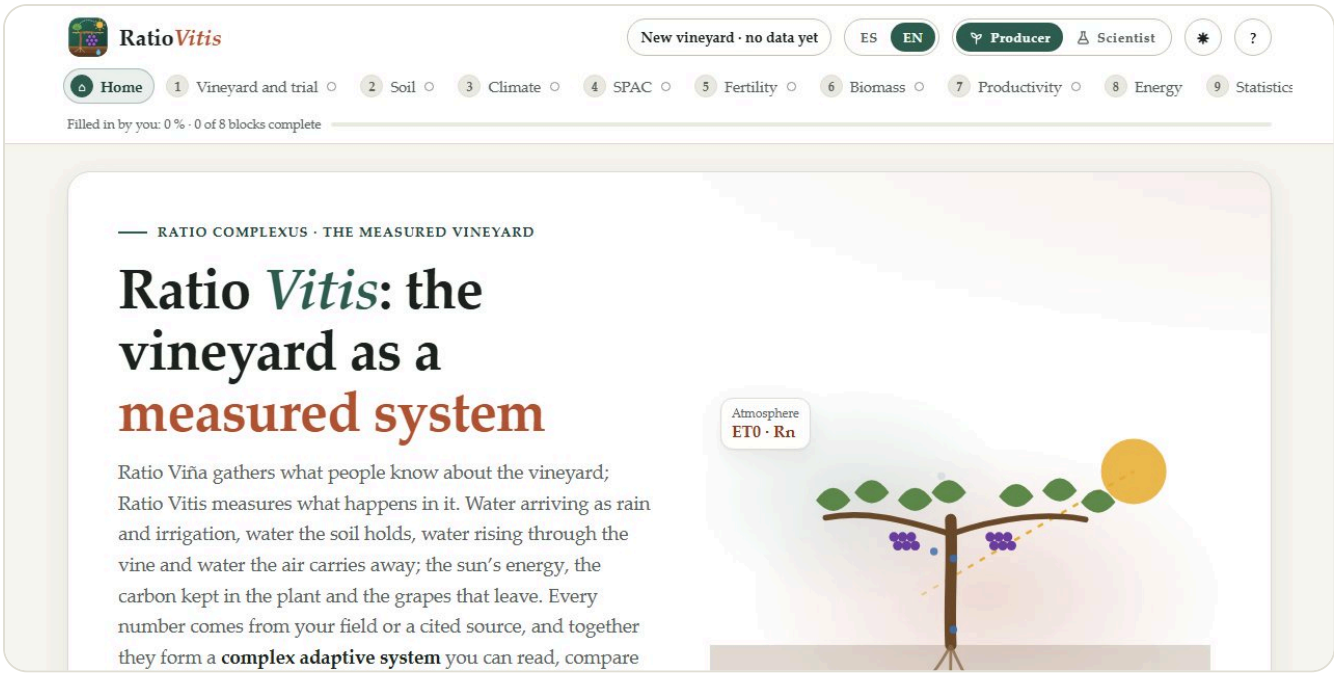


FIGURE 0.4 **Entering the example.** The link on Home.

1 View the Mata de Zarza example.



FIGURE 0.5 **Example mode.** The band and each block's notice.

1 Orange band with the button to leave. 2 "Example · these data are not yours".

Inside the example, Home tells you where each datum comes from:

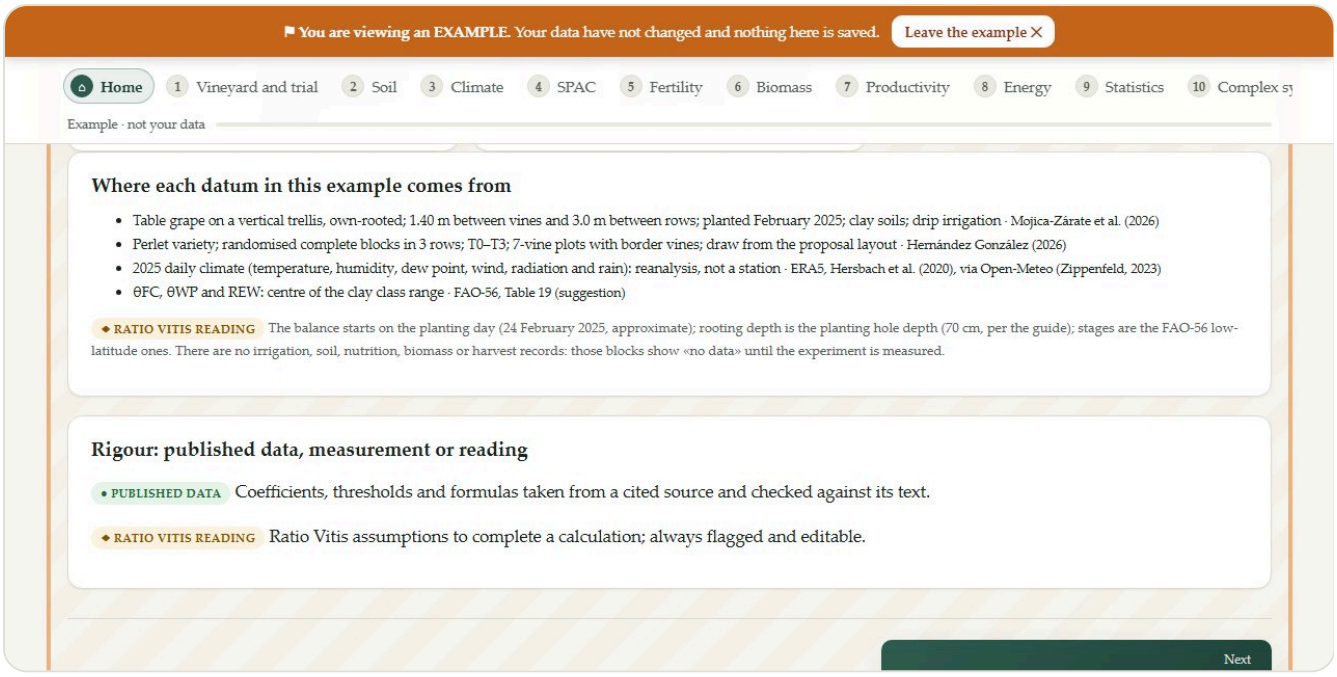


FIGURE 0.6 **Example sources.** Establishment guide, thesis proposal, ERA5 reanalysis and FAO-56; below, the flagged readings.

> **!** **THE EXAMPLE CLIMATE IS A REANALYSIS**
>
> Mata de Zarza has no weather station of its own. The 2025 daily climate comes from the ERA5 reanalysis (Hersbach et al., 2020), downloaded from Open-Meteo for the 19.25° N, 96.75° W cell adjusted to 467 m. It is fine for learning, but for your thesis or irrigation use records from a nearby station or your own.

BLOCK 1 · VINEYARD AND TRIAL

1 Where it is, how it is planted and what you will compare

The site gives latitude and altitude; the vineyard, the water coefficients; the design, a fair comparison between treatments.

1.1 The site

Choose the INEGI state, municipality and locality; from them the app takes the latitude and altitude it needs to compute solar radiation and air pressure. If your vineyard is far from the locality, tick **Enter measured latitude and altitude (GPS)** and type them in.



FIGURE 1.1 **1.1 Site.** Mata de Zarza, Puente Nacional, Veracruz: 19.230° N and 467 m.

1.2 The vineyard

Choose whether the grape is **table or raisin** or **wine**: that sets the FAO-56 coefficients the water balance uses. Enter the variety, rootstock, planting year, row and vine spacing, training system and irrigation type. The app works out vines per hectare.

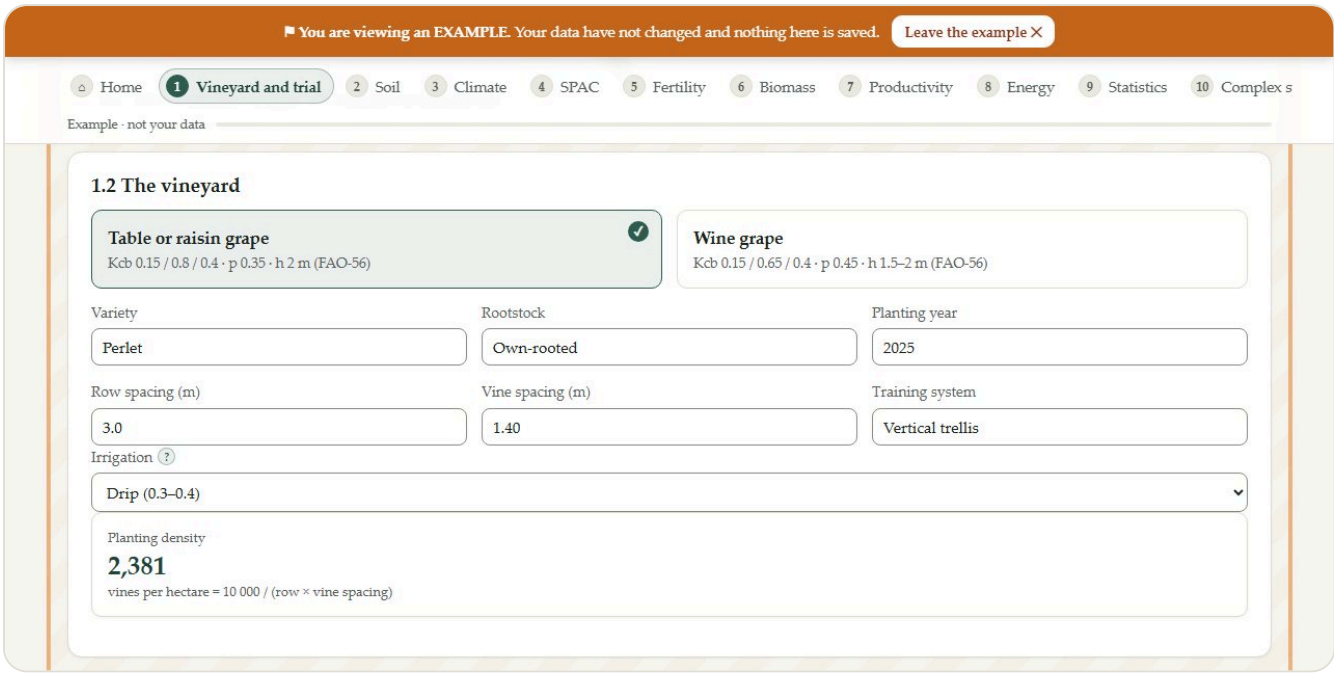


FIGURE 1.2 **1.2 The Mata de Zarza vineyard.** 3.0 m × 1.40 m give 2,381 vines per hectare.

✓ **VINES PER HECTARE**

Divide 10,000 m² by row spacing times vine spacing: 10,000 ÷ (3.0 × 1.40) = **2381**.

1.3 Trial design and draw

If you will compare treatments, enter them here (always include a **control**), say how many blocks there are, how many vines per plot and how many border vines you leave at each end. **Draw again** assigns treatments at random within each block; the same seed always gives the same layout. Download the **Field sheet**: it is where you will record each vine.

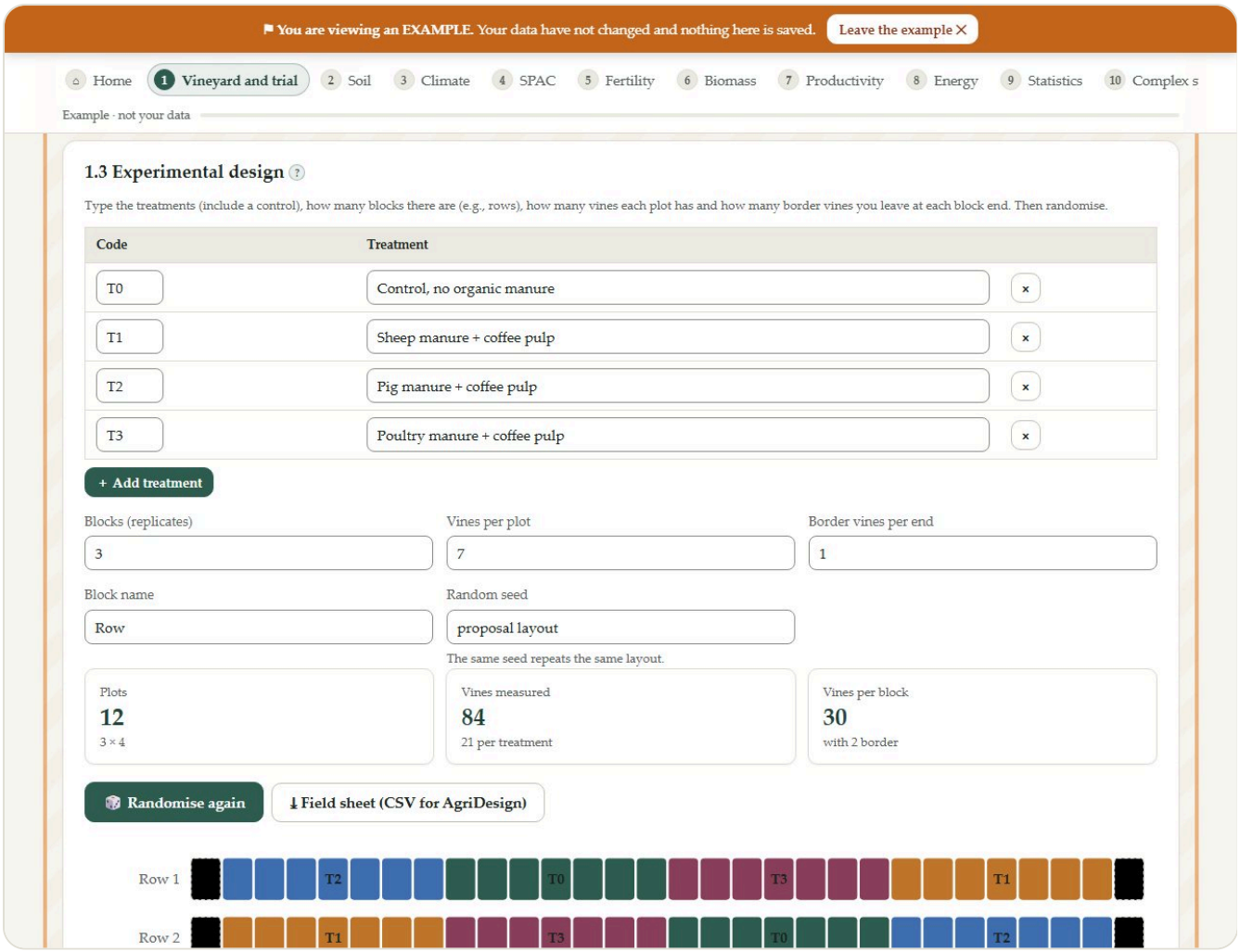


FIGURE 1.3 **1.3 The Mata de Zarza trial.** Four treatments in three rows: 12 plots and 84 vines measured.

IN MATA DE ZARZA

Each row of about 30 vines is a block with the four treatments: **T0** control without manure, **T1** sheep manure + coffee pulp, **T2** pig + pulp and **T3** poultry + pulp. Plots of 7 vines and one border vine at each end. The example uses exactly the draw from the thesis proposal layout (Hernández González, 2026).

WHY NOT ONE MANURE PER ROW?

If each row gets a single manure, the manure effect is mixed with the row effect (better soil, more shade) and there are no true replicates. With randomised blocks each treatment is repeated in all three rows.

1.4 Study variables

Each case study measures what it can. Here you switch off the variables you will not measure: they disappear from the blocks and from the progress counts, and nothing is computed with them. Phenological dates are switched on or off here too.

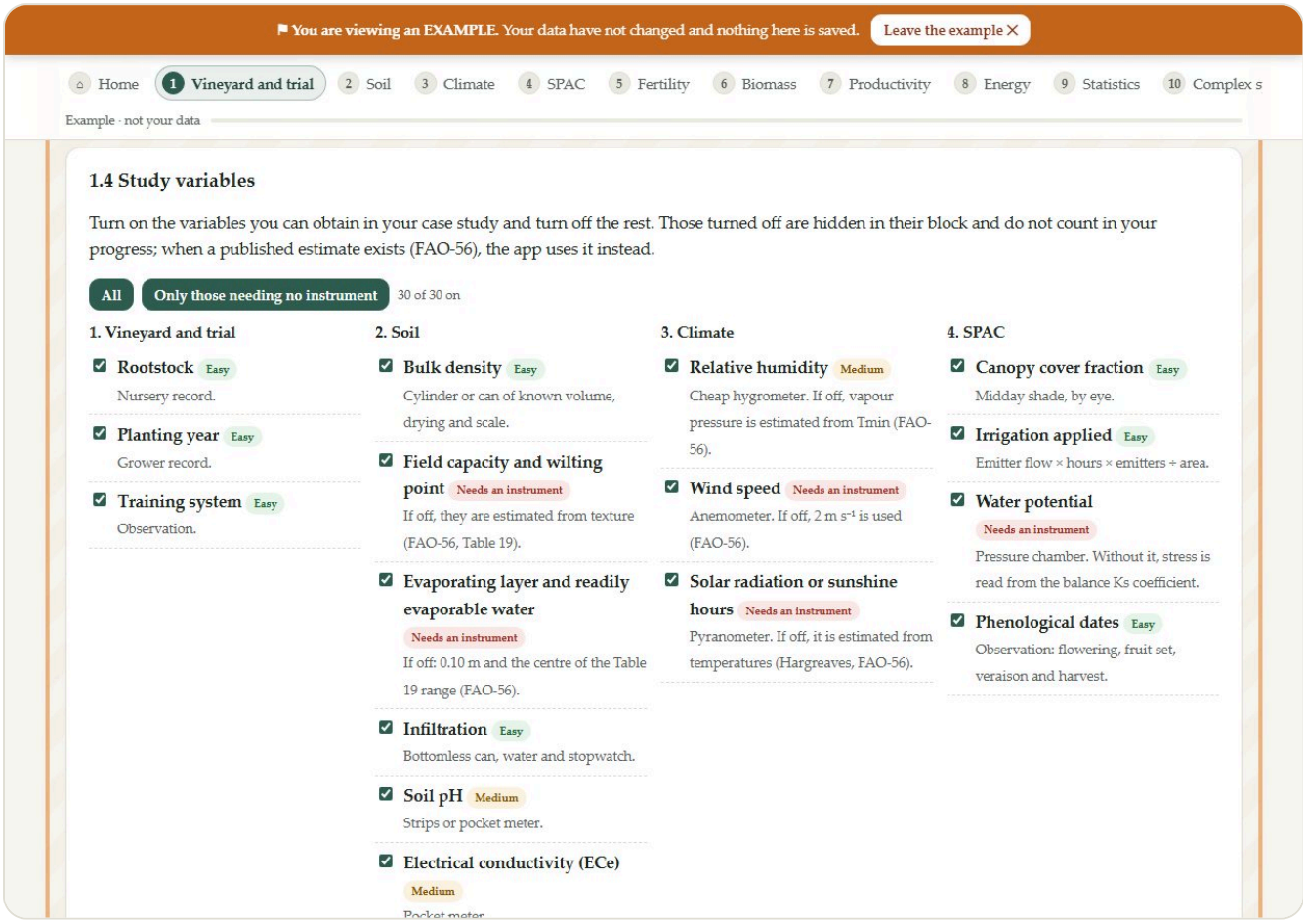


FIGURE 1.4 **1.4 Study variables.** Each says how hard it is to measure: easy, medium or needs an instrument.

BLOCK 2 · SOIL

2 The vine's water tank

With texture and rooting depth you already know how much water your soil holds; pH and salt tell you whether anything holds the roots back.

2.1 Texture: the jar test

Fill a third of a straight glass jar with dry, sieved soil from the root zone, add water almost to the top and a teaspoon of dish soap, close and shake for several minutes. Measure the sand height after a minute, silt after two hours and clay when the water clears (one or two days). Enter the three heights: the app gives the percentages and the USDA texture class.

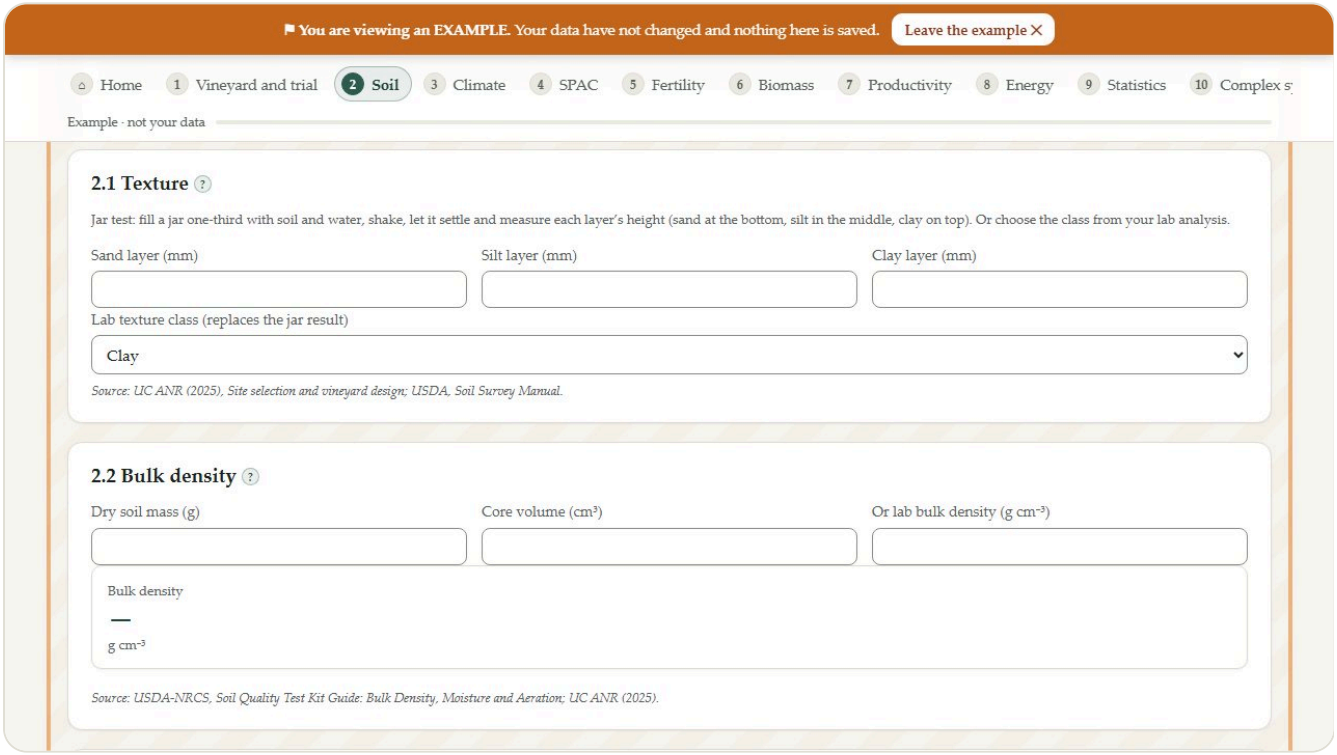


FIGURE 2.1 **2.1 and 2.2 Texture and bulk density.** The example has only the "Clay" class from the guide.

2.2 Bulk density

Drive in a cylinder of known volume (a bottomless can will do), lift it full without compacting it, dry the soil in the sun or an oven and weigh it. Dry mass over volume gives bulk density; the traffic light says whether it already restricts roots for your texture (USDA-NRCS).

2.3 Available water

The water the vine can use lies between **field capacity** and **wilting point**. If you have those values from the laboratory, enter them; if not, the app suggests the centre of your texture's range (FAO-56, Table 19), always flagged as suggested. With the **rooting depth** it computes total available water (TAW) and readily available water (RAW).

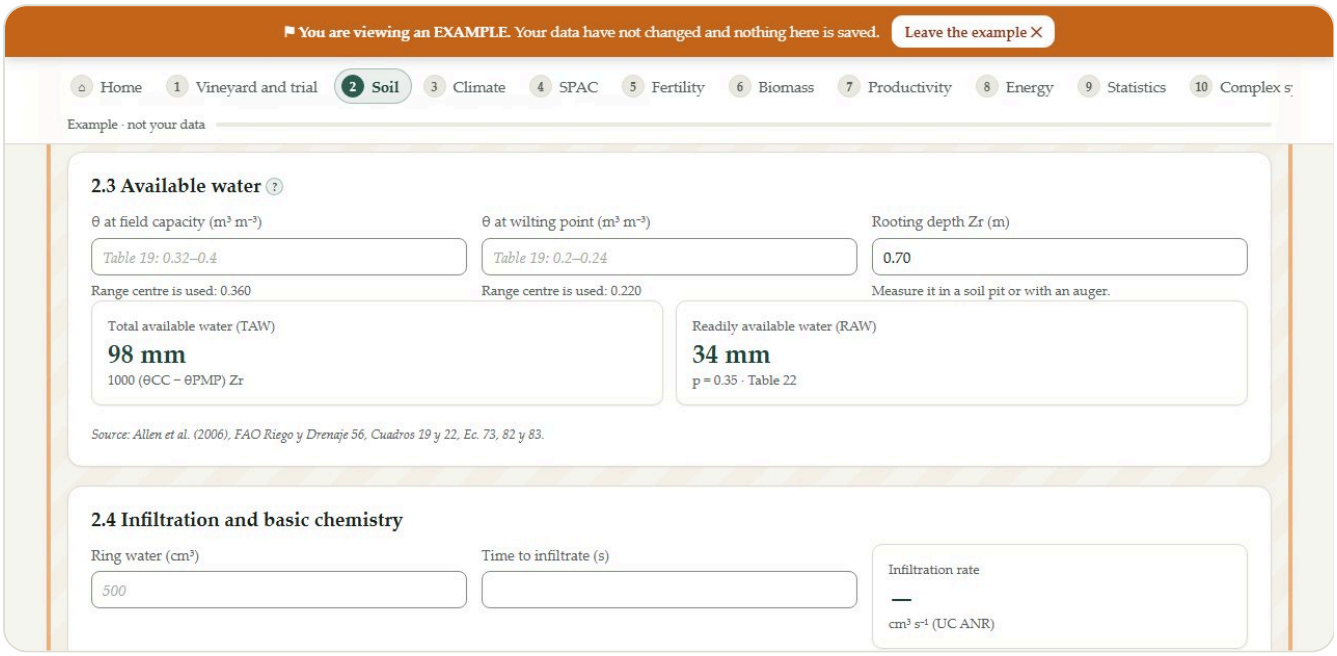


FIGURE 2.2 **2.3 Available water in Mata de Zarza.** 98 mm in total, 34 mm easy to take up.

◆ **IN MATA DE ZARZA**

The guide describes the area as having **clay soils**; with that class the app suggests θFC and θWP from Table 19. The rooting depth of **0.70** m is the planting-hole depth (70 cm) and is flagged as a Ratio Vitis reading. With that: TAW = 98 mm and RAW = 34 mm. When the soil analysis arrives, its values replace the suggested ones.

2.4 Infiltration, pH, salt and organic matter

With a buried tube and a depth of water you measure how long it takes to infiltrate. pH, electrical conductivity and organic matter come from the laboratory. The app classes pH for the vine (ideal 5.5–6.5) and estimates the yield lost to salt (FAO-56: threshold of 1.5 dS m^{-1} and 9.6 % less per dS m^{-1} above).

→ **IF THE SOIL TRAFFIC LIGHT SHOWS…**

Compaction	Loosen before fertilising: roots cannot use what they cannot reach.
pH outside 5.5–6.5	Confirm with another sample; correct with lime or organic matter as appropriate.
EC above 1.5 dS m^{-1}	Check irrigation water and salt leaching.

BLOCK 3 · CLIMATE

Daily records, ET0 and viticultural indices

3

The air demands water and the sun supplies energy. From your station's daily records the app computes each day's demand and how warm your region is for grapes.

3.1 Daily records

Import a CSV file with one row per day. Required columns are **date**, **maximum** and **minimum temperature**; if your station gives humidity, wind, radiation or rain, include them and the calculation improves. The app recognises headers in Spanish or English and tells you what it estimated with FAO-56 when something is missing. **CSV for PhenologyPro** exports the same data for Luis Ángel Barrera-Guzmán's app.

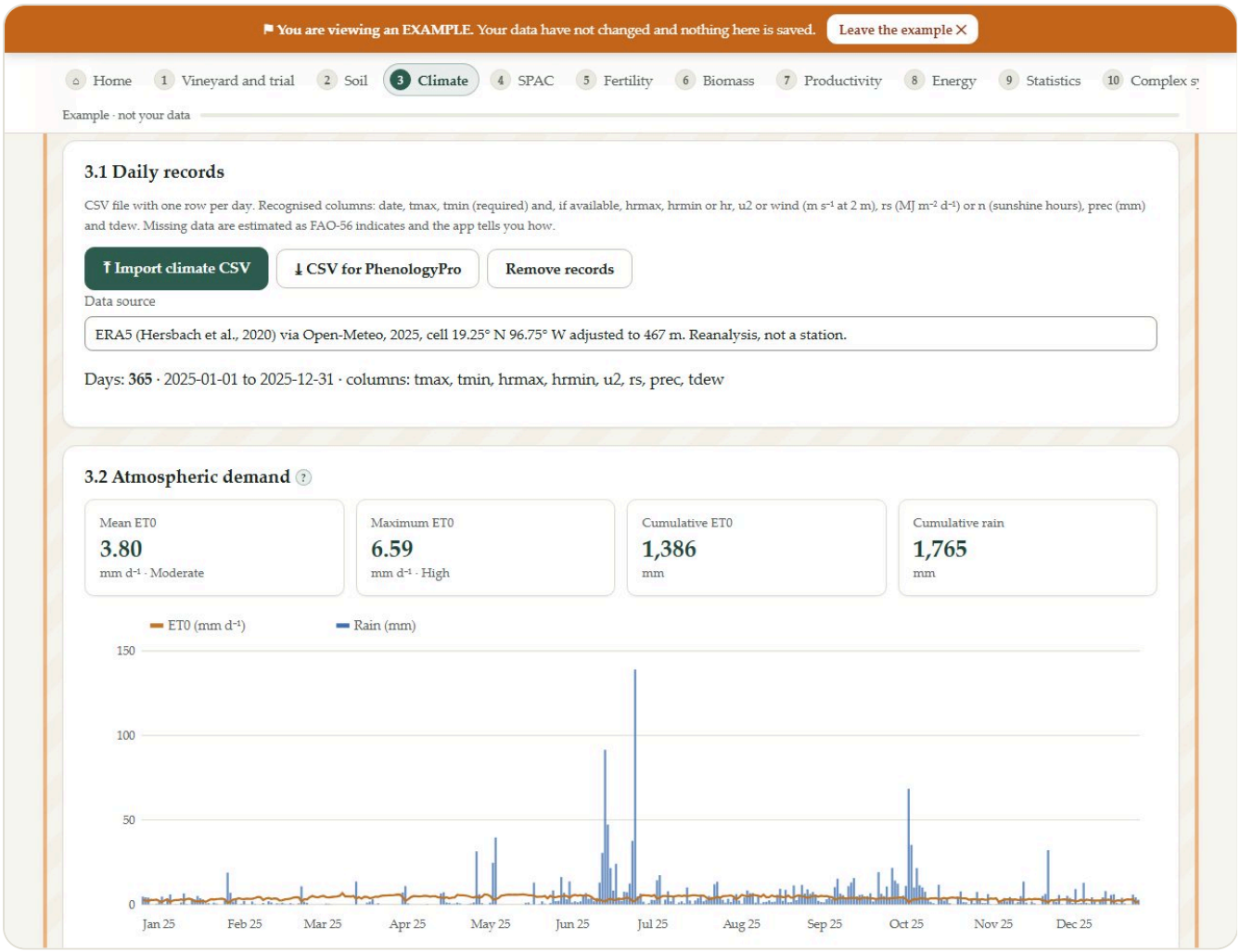


FIGURE 3.1 **3.1 and 3.2 Mata de Zarza climate, 2025.** 365 days of ERA5; mean ET0 of 3.80 mm a day and 1,765 mm of rain.

3.2 What the air demands

ET0 is the water the air demands from a well-watered grass in a day; the app computes it with FAO Penman–Monteith. 1 to 3 mm a day is low demand; 3 to 5, moderate; 5 to 7, high; above 7, very high (FAO-56). The chart shows each day's rain beside it.

3.3 Viticultural indices

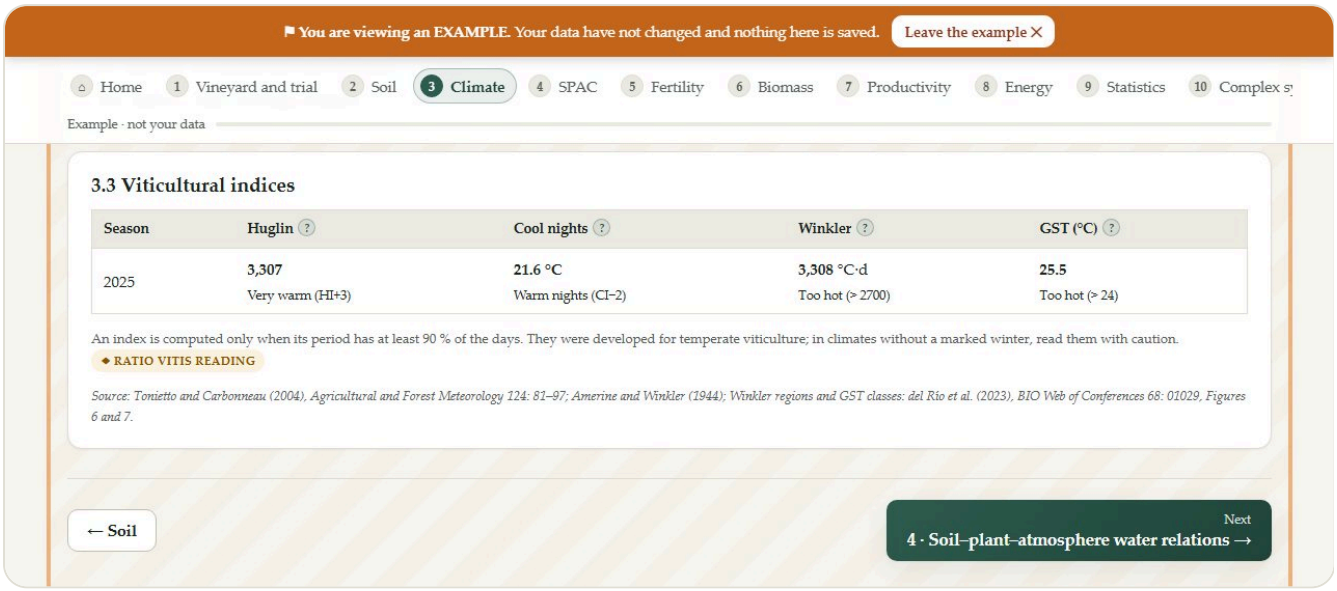


FIGURE 3.2 **3.3 Indices for 2025 in Mata de Zarza.** All fall in the warmest class.

INDEX	WHAT IT MEASURES	MATA DE ZARZA 2025	SOURCE
Winkler	Useful heat 1 Apr–31 Oct, above 10 °C. Regions I (850–1389) to V (2223–2700) °C·d.	3,308 °C·d · too hot (> 2700)	del Río et al. (2023), Fig. 6
GST	Mean temperature 1 Apr–31 Oct.	25.5 °C · too hot (> 24)	del Río et al. (2023), Fig. 7
Huglin	April–September heat weighting sunny hours.	3,307 · very warm (HI+3)	Tonietto y Carbonneau (2004)
Cool nights	Mean September minimum.	21.6 °C · warm nights (CI−2)	Tonietto y Carbonneau (2004)

! **INDICES BUILT FOR TEMPERATE CLIMATES**

These indices were born in viticulture with a marked winter. In a warm sub-humid climate like Mata de Zarza's, where the vine barely rests, they read "too hot" for the April–October calendar, but they do not say grapes cannot be grown: the guide documents their establishment. Read them as a comparison with other regions, not as a verdict.

BLOCK 4 · SPAC

Soil–plant–atmosphere water relations

4

Water enters as rain and irrigation, is stored in the soil, rises through the vine and leaves to the air. Here it is followed day by day with the FAO-56 dual coefficient.

4.1 Stages and budburst

Choose the FAO-56 Table 11 stage lengths closest to your region and enter the **budburst date**: the water balance runs from that day.

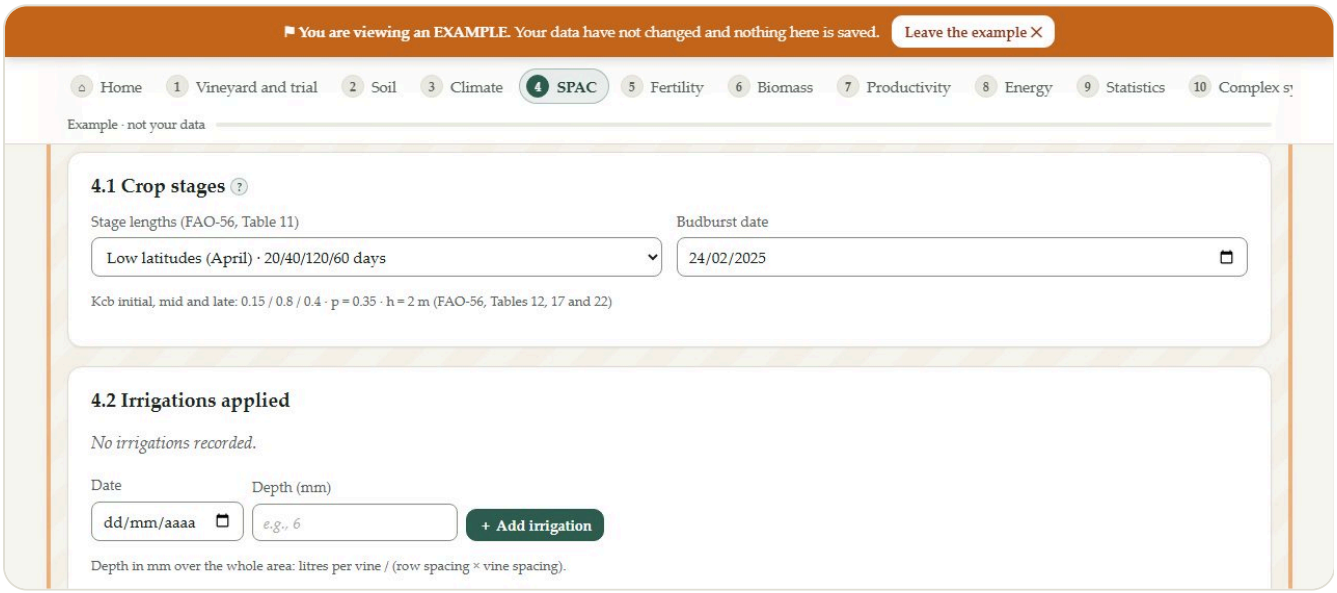


FIGURE 4.1 **4.1 Stages in Mata de Zarza.** Low latitudes (20/40/120/60 days) from planting.

4.2 Irrigations applied

Record each irrigation with its date and depth in millimetres. If you measure litres per vine, divide by each vine's area: litres ÷ (row spacing × vine spacing) = mm. In Mata de Zarza, 10 litres per vine is 10 ÷ (3.0 × 1.40) = 2.4 mm.

4.3 The daily balance

Everything comes together here: the air demands (Block 3), the soil stores (Block 2) and the vine drinks. Each day the app subtracts what is evapotranspired and adds rain and irrigation. **Depletion** says how much water the soil lacks; when it exceeds RAW, the **stress coefficient Ks** drops below 1 and the vine starts closing its stomata.

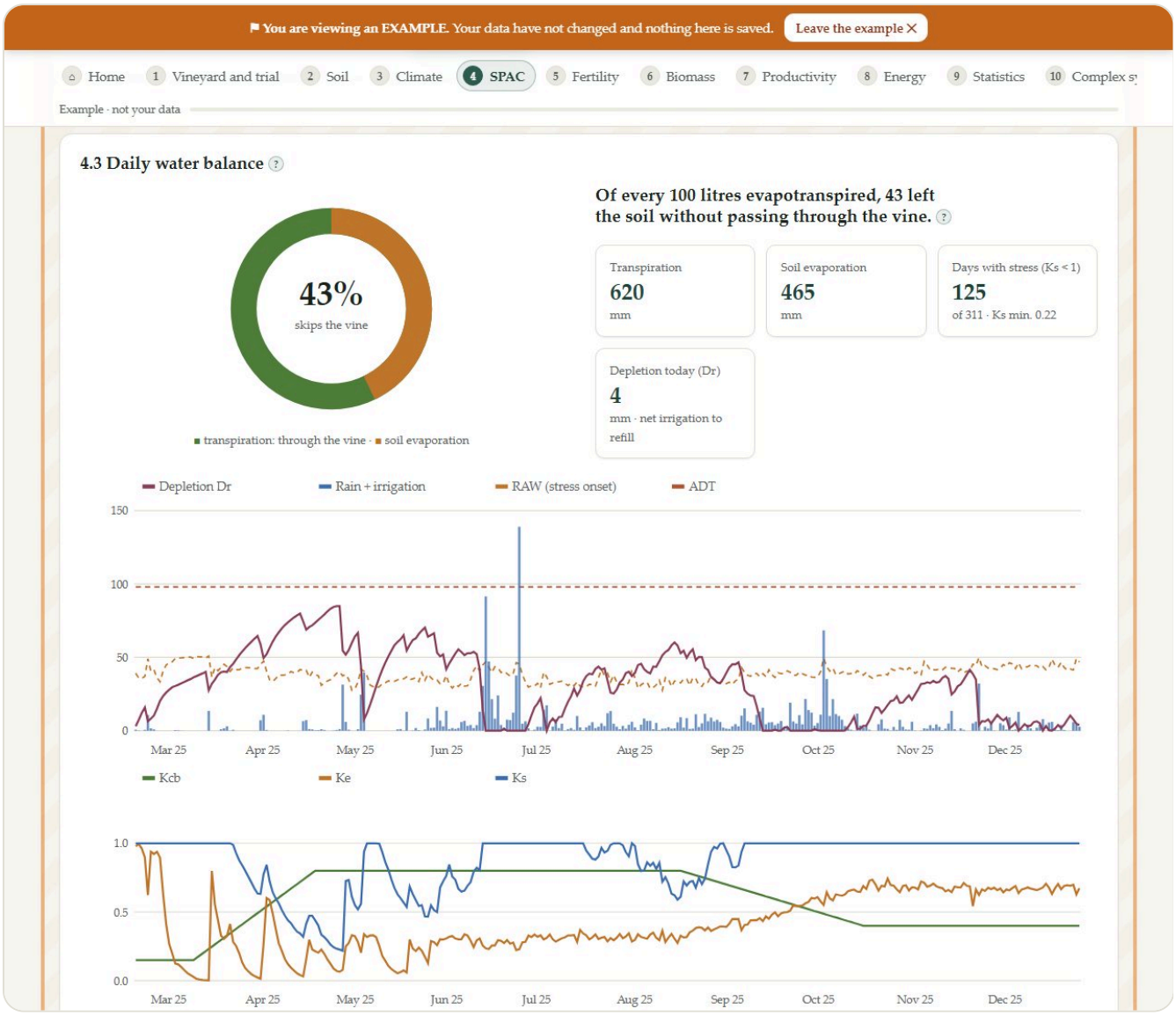


FIGURE 4.2 **4.3 Mata de Zarza balance, 2025, with no irrigation recorded.** 43 of every 100 litres left the soil without passing through the vine.

IN MATA DE ZARZA

From 24 February 2025, 1,085 mm were evapotranspired: **620** as vine transpiration and **465** evaporated from the soil. There were **125 stress days** out of 311, with a minimum Ks of 0.22, mostly in the dry season before the June rains. Because no irrigation is recorded, the balance shows the vine as if rainfed; the guide calls for constant irrigation in the first three months, so in reality spring stress must have been lower. Recording each irrigation is what makes the balance faithful.

! **A YOUNG VINE TRANSPIRES LESS**

FAO-56 coefficients are for a mature vine. With one- or two-year-old vines, enter the ground-cover fraction (fc) in the scientist view or measure NDVI in Block 6; otherwise transpiration comes out high.

→ **WHEN TO IRRIGATE**

Ks = 1	No stress: no need to irrigate yet.
Ks < 1	Irrigate the "Depletion today (Dr)" depth and record it in 4.2.
Much water below the root zone	You over-irrigate or it rained hard: space out irrigations.

4.4 Water potential and phenology

If you have a pressure chamber, measure **midday stem** potential (leaf bagged an hour before) or **predawn leaf** potential. The app classes it on four published scales (IVES, UC ANR and Uriarte et al., 2025). If your chamber reads bar, divide by 10. In 4.5 record the date on which half the vines of each plot reached flowering, fruit set, veraison and harvest.

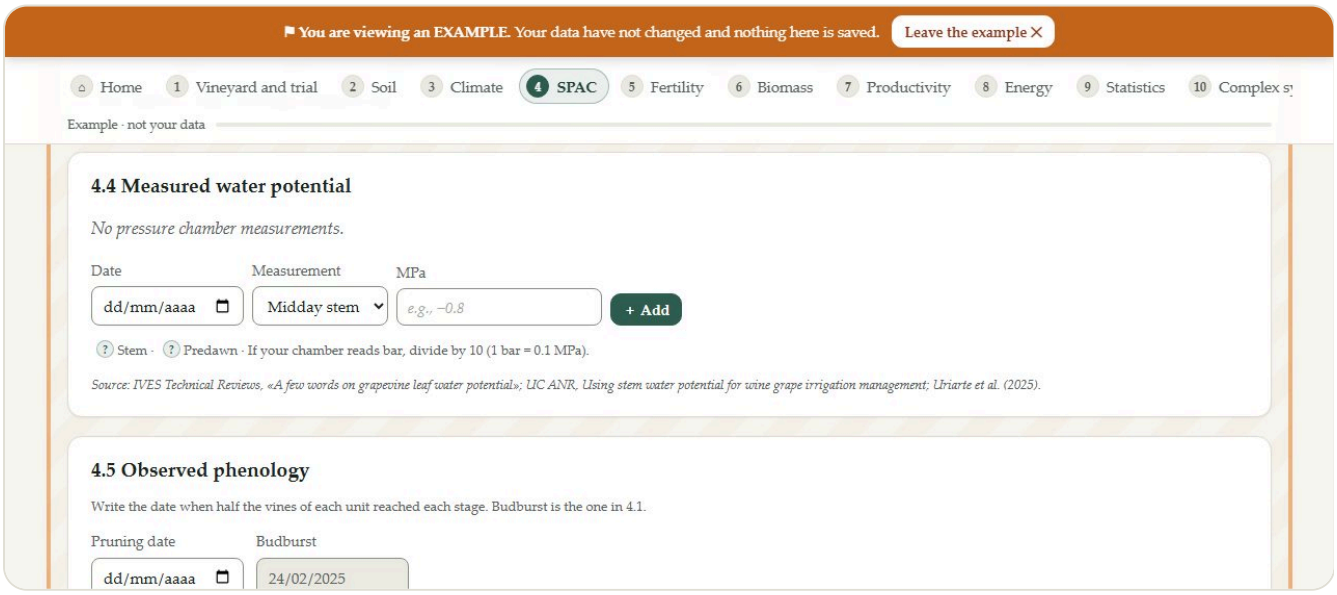


FIGURE 4.3 **4.4 and 4.5 Water potential and phenology.** No measurements yet in the example.

BLOCK 5 · FERTILITY

5 What the soil offers and what the vine took up

The soil analysis says what is there; the leaf or petiole analysis says what reached the plant.

5.1 Soil analysis

Enter the laboratory's total nitrogen, Olsen phosphorus, exchangeable bases and CEC. They are used to compute the manure dose and the sustainability index.

5.2 Tissue analysis and DOP

Choose the tissue (petiole or blade) and the time (flowering, with the leaf opposite the cluster, or veraison) and enter each nutrient's concentration per treatment. The app reads them against published sufficiency ranges and with the **DOP index**: negative, the nutrient is lacking; positive, in excess; the most negative is the most limiting, and the sum of all (Σ|DOP|) measures imbalance.

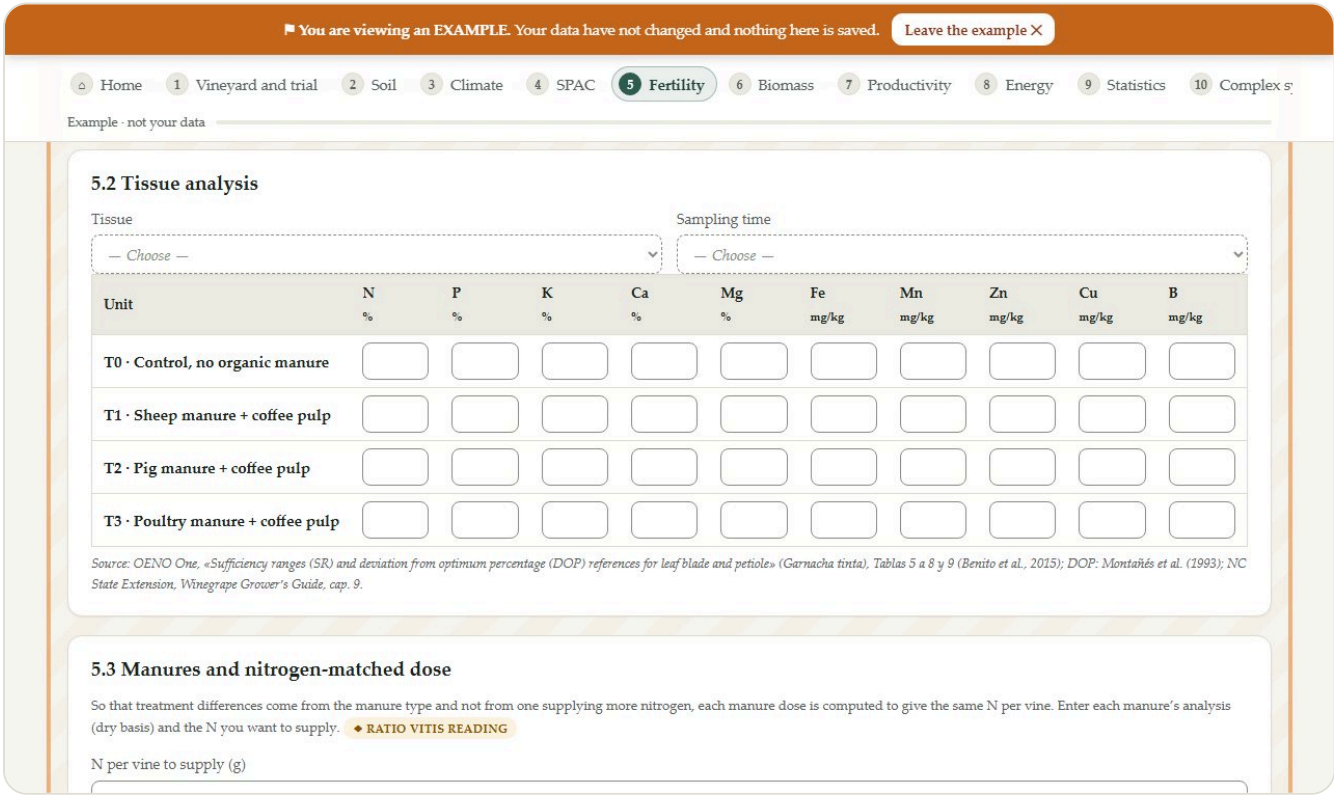


FIGURE 5.1 **5.2 Tissue analysis.** One row per treatment.

! **REFERENCES FROM ANOTHER VARIETY**

The ranges and DOP references are for red Grenache in La Rioja (Benito et al., 2015). They are useful for comparison, not standards for Perlet or Mexico. A low nutrient is confirmed with the soil analysis before correcting.

5.3 Manures and nitrogen-matched dose

So that the difference between treatments comes from the type of manure and not from one supplying more nitrogen, the app computes each manure dose to give the same N per vine. Enter each mixture's analysis (N, carbon, moisture) and the N you want to supply.

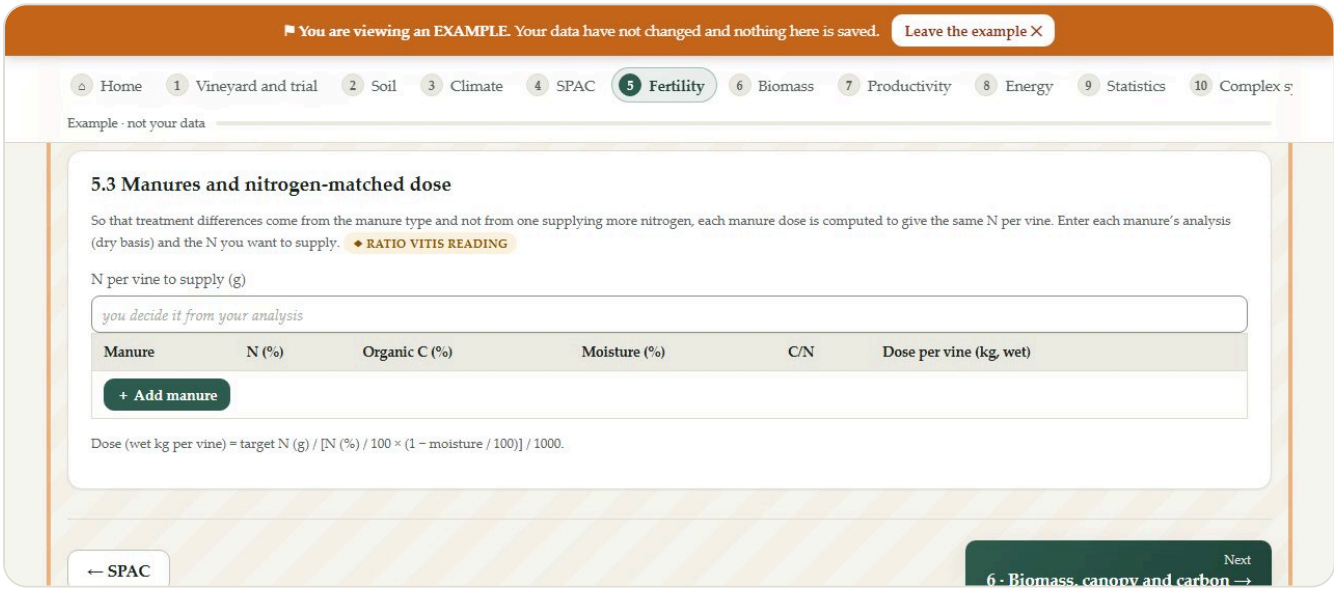


FIGURE 5.2 **5.3 Nitrogen-matched dose.**

✓ **COMPOST BEFORE APPLYING**

The thesis proposal recommends composting each manure–coffee pulp mixture at an initial C/N near 25–30 and analysing it before application.

BLOCK 6 · BIOMASS

6 Vigour, canopy and carbon

Vigour shows in the prunings and the leaves; dry matter says how much carbon the vine stored.

6.1 Vigour and canopy

Weigh the winter prunings of the tagged vines in each plot: it is the cheapest and most useful datum of the block. If you can, measure leaf area and NDVI; from NDVI the app estimates the ground cover fraction (Er-Raki et al., 2013), which corrects the water balance of a young vine.

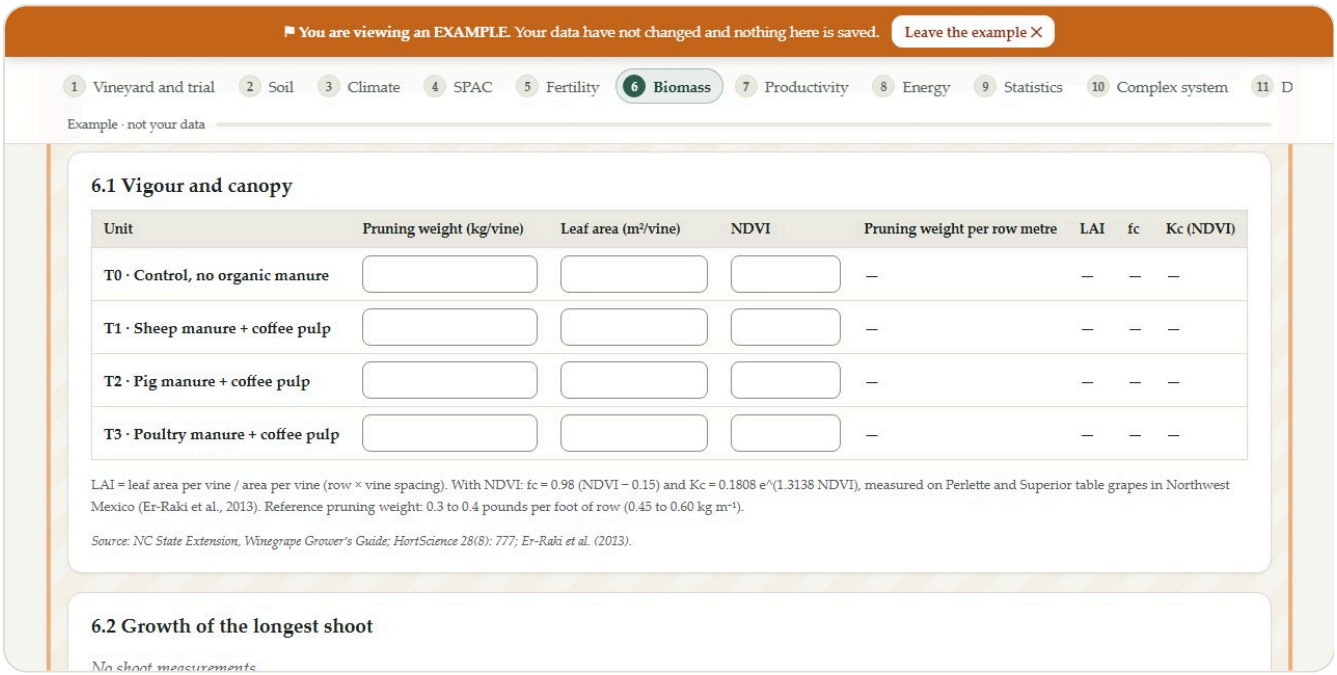


FIGURE 6.1 **6.1 Vigour and canopy.** Pruning weight, leaf area and NDVI per treatment.

6.2 Shoot growth and carbon

Measure each vine's longest shoot on several dates to see its growth rate. At the start and end of the cycle, the thesis samples vines to weigh root, stem, leaf and fruit dry matter; with the carbon fraction the app computes accumulated carbon, fixed CO_2 (×44/12) and released O_2 (×32/44).

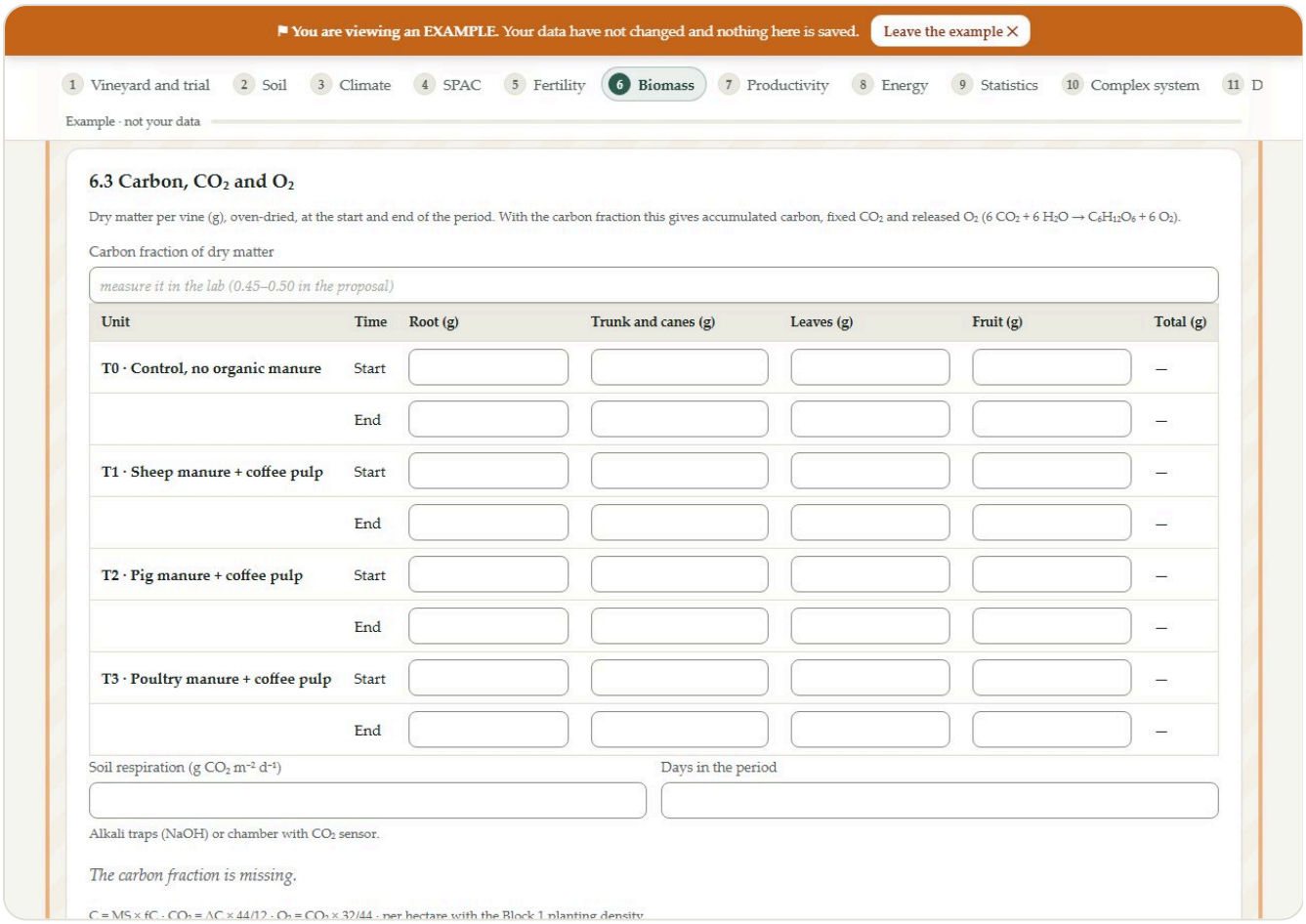


FIGURE 6.2 **6.3 Carbon, CO_2 and O_2.** Initial and final dry matter per treatment.

BLOCK 7 · PRODUCTIVITY

7 How much fruit, of what quality and with how much water

What leaves the vineyard and what it cost in water.

7.1 Harvest and quality

Harvest the same tagged vines of each plot and record on the same day yield per vine, clusters, berry weight and °Brix. The app computes tonnes per hectare, cluster weight, Ravaz index (with Block 6 prunings), leaf area/fruit ratio and water productivity.

7.2 How much is lost to water shortage

From the Block 4 balance, the app estimates what share of maximum yield is lost to water shortage alone, with the grape Ky factor (0.85, FAO-56, Table 24).

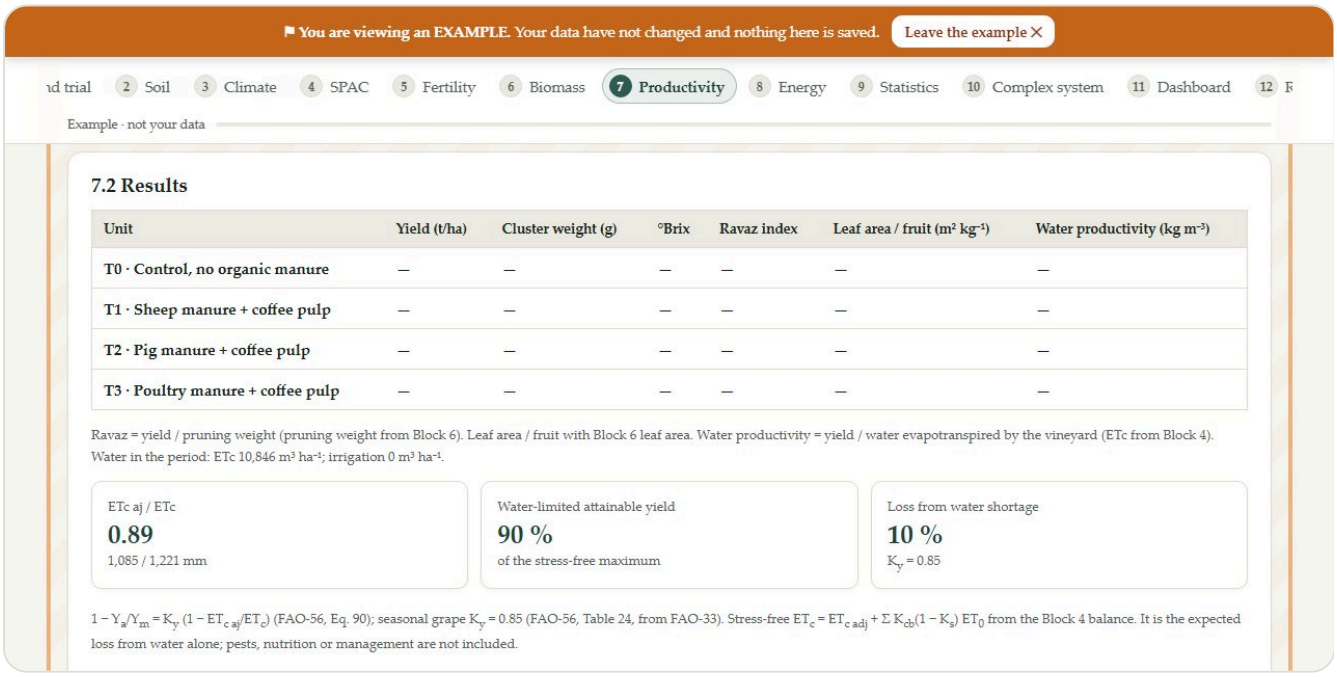


FIGURE 7.1 **7.2 Results in Mata de Zarza.** No harvest measured yet, but with the expected loss from water: 10 %.

◆ **IN MATA DE ZARZA**

In 2025 the vine evapotranspired 89 % of what it would have used without stress (1,085 of 1,221 mm). With Ky = 0.85, water-limited attainable yield is **90 %** of maximum: water shortage alone would cost about **10 %**. Pests, nutrition and management are not part of this figure.

→ **HOW TO READ THE HARVEST**

Ravaz < 5	Much vigour for the fruit: check nitrogen and irrigation.
Ravaz > 10	Overcropped vine: thin clusters.
°Brix 14 to 17.5 in table grapes	Within the harvest window (Permanhani et al., 2016).

BLOCK 8 · ENERGY

8 How the vineyard shares out radiation

No new entries: this block is computed from climate and the water balance.

The sun delivers energy to the vineyard. The part spent evaporating water (latent heat) cools it; the rest heats the air (sensible heat). The app splits each day's net radiation using the Block 4 water balance and computes the Bowen ratio: the lower it is, the more energy goes into water.



FIGURE 8.1 **8.1 Energy in Mata de Zarza, 2025.** 3,755 MJ m^{-2} of net radiation: 2,657 as latent heat and 1,098 as sensible heat; Bowen of 0.41.

? **WHY DOES IT MATTER?**

A well-watered vineyard spends more energy cooling itself; when water runs short, Bowen rises and leaves heat up. It is another way to see Block 4 stress.

BLOCK 9 · STATISTICS

9 Is the difference real or could it be chance?

Statistics compares treatments taking into account variation between rows and between vines.

Once you have field data, fill in the sheet you downloaded in Block 1 (one row per plot or vine, with treatment, block, date and variable columns) and import it here. Choose which column is the treatment, which the block and which variable to analyse. The app does descriptive statistics, the randomised-block ANOVA, Tukey's test, ANCOVA with vine age and repeated measures over time.

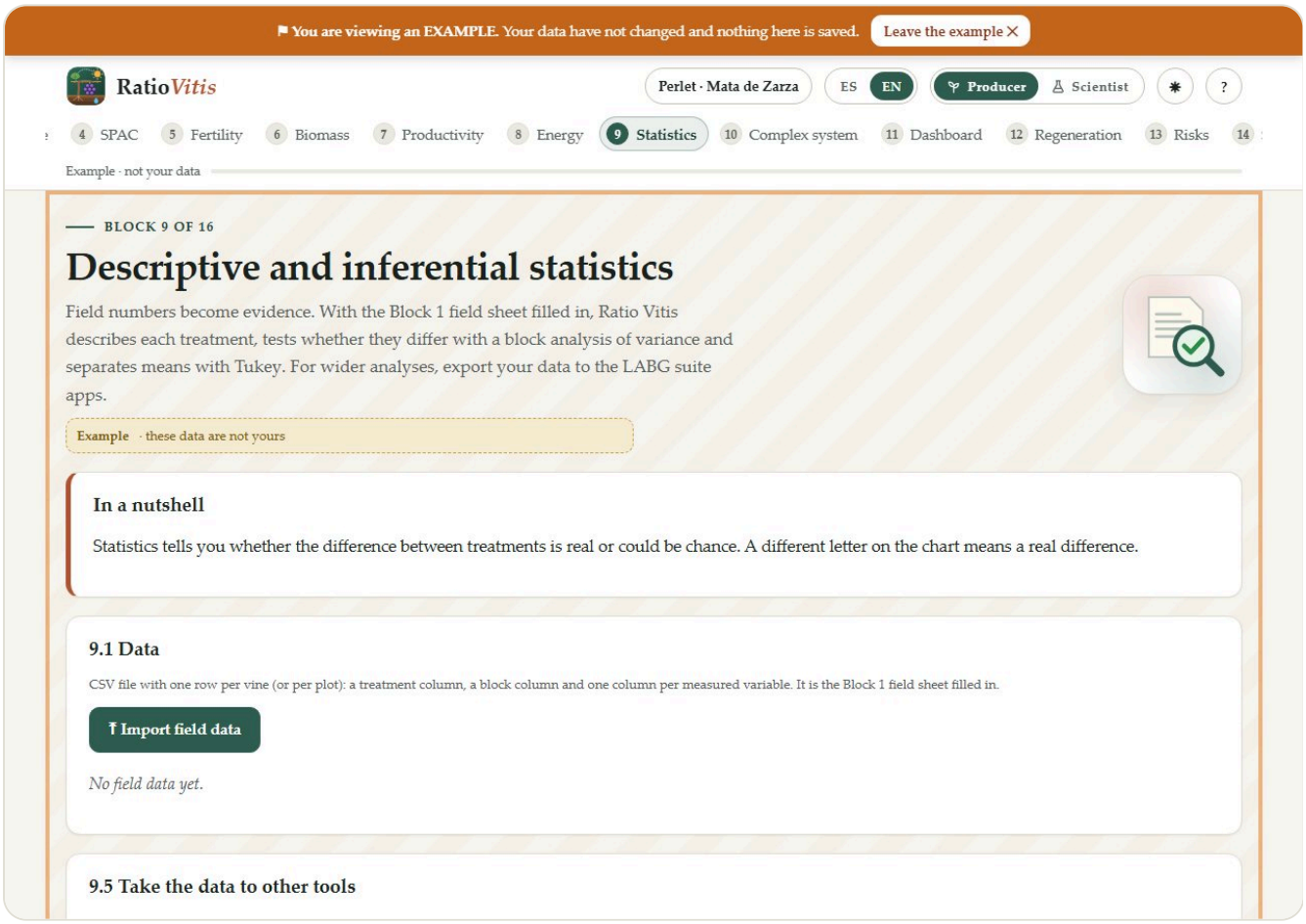


FIGURE 9.1 **Block 9.** Without field data, only the import and the links to other tools appear.

→ **HOW TO READ THE RESULT**

$p < 0.05$ in the ANOVA	At least one treatment differs; look at Tukey's letters.
Different letters	Real difference between those two treatments.
Same letter	The difference could be chance.

✓ **TAKE IT TO THE LABG SUITE APPS**

From here you export the data in the format of AgriDesign, CorrelationPro and PCAPro, by Luis Ángel Barrera-Guzmán, for charts and multivariate analysis.

BLOCK 10 · COMPLEX SYSTEM

10 The vineyard as a complex adaptive system

Measurements join into a network with feedback loops.

Your vineyard is a network: what you change in the soil reaches the leaf and comes back. Block 10 joins the variables of blocks 1 to 9 into a network with their direction (more soil water → more transpiration) and tells it to you in a **Walkthrough** of eight scenes with your own numbers.

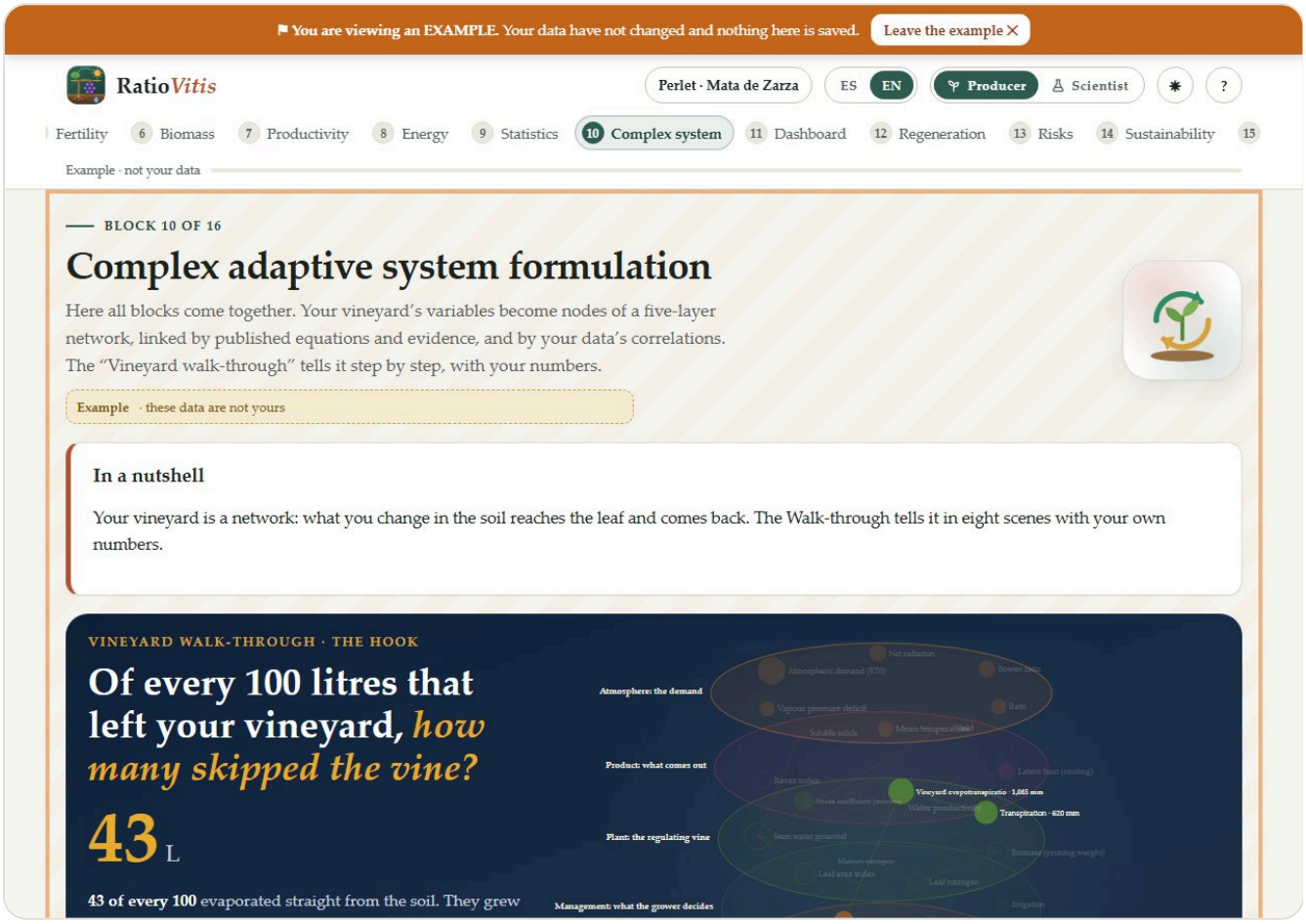


FIGURE 10.1 **The vineyard Walkthrough.** The first scene with Mata de Zarza data: 43 of every 100 litres did not pass through the vine.

10.1 Feedback loops

A **loop** is a path that returns to its start. If it **balances** (negative), the system stabilises itself: for example, the soil dries, the vine closes stomata, transpires less and the soil dries more slowly. If it **reinforces** (positive), a small change grows.

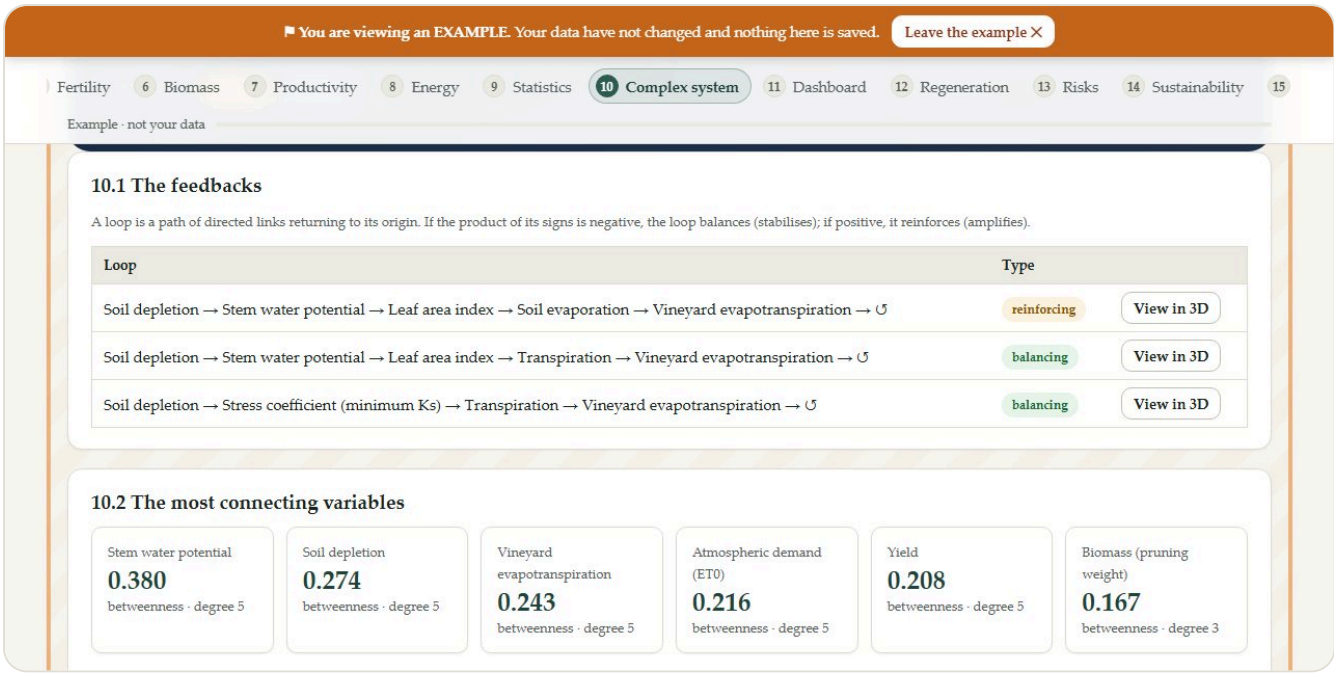


FIGURE 10.2 **10.1 and 10.2 Loops and most-connected variables.** Soil depletion and stem potential are the system's bridges.

BLOCK 11 · DASHBOARD

11 The whole vineyard on one page

A summary built only from what you entered.

Your whole vineyard on one page: key figures, a traffic light with each indicator's source class, where the water went, the comparison between treatments and what you have entered so far. Start with the traffic light: red and amber come first.

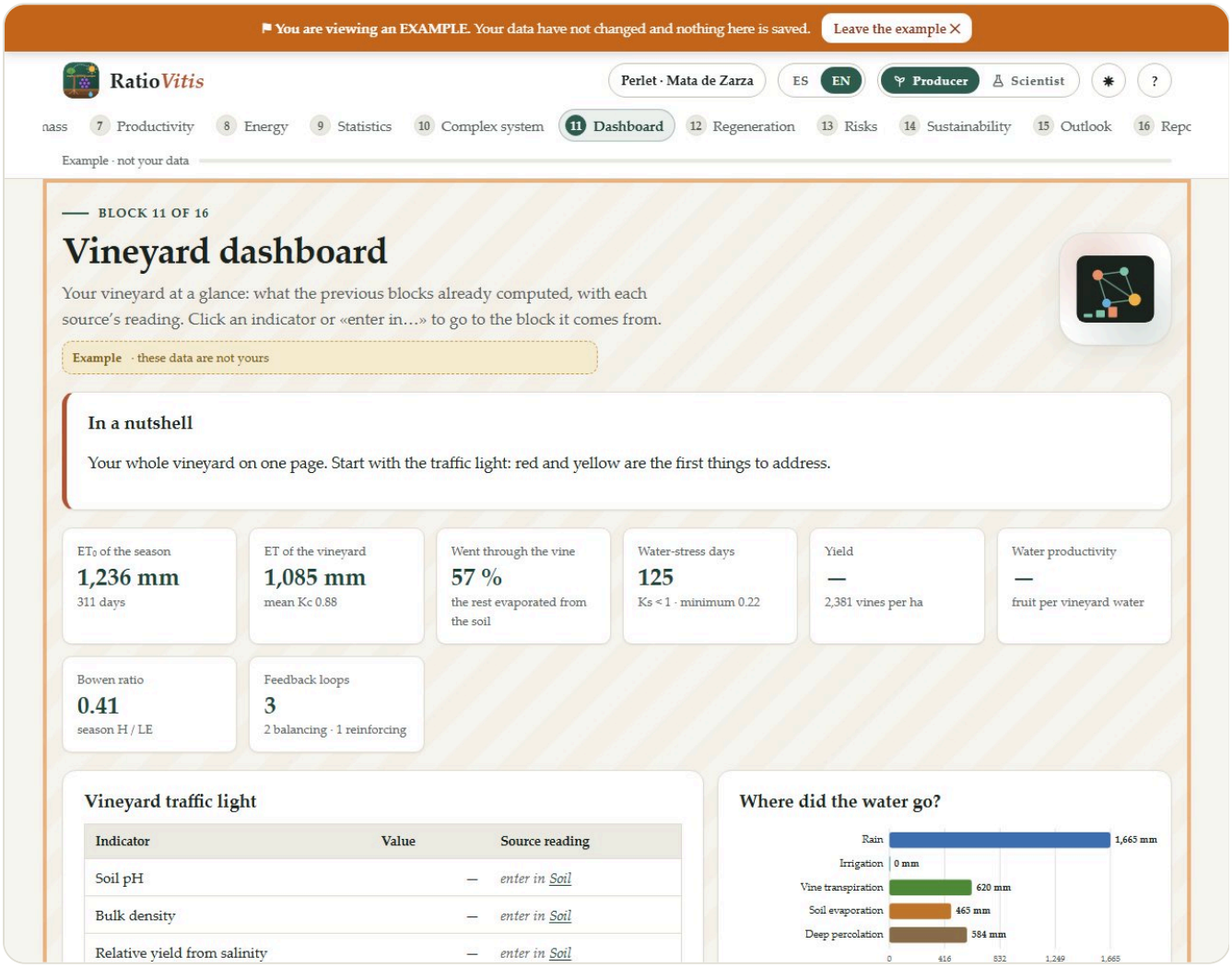


FIGURE 11.1 **Mata de Zarza dashboard.** Season ET0 of 1,236 mm and 57 % of the water through the vine.

BLOCK 12 · REGENERATION

12 Regenerating or wearing out?

Practices, outcomes and water resilience.

Regenerative agriculture seeks for each cycle to leave soil, water and biodiversity better than the last. Score your practices 0 to 4 (processes) and enter the outcomes you measure (organic matter, carbon, water); save a **baseline** to compare next cycle. Below, water resilience: how many days the soil lasts without irrigation and how much stress held the vine back.

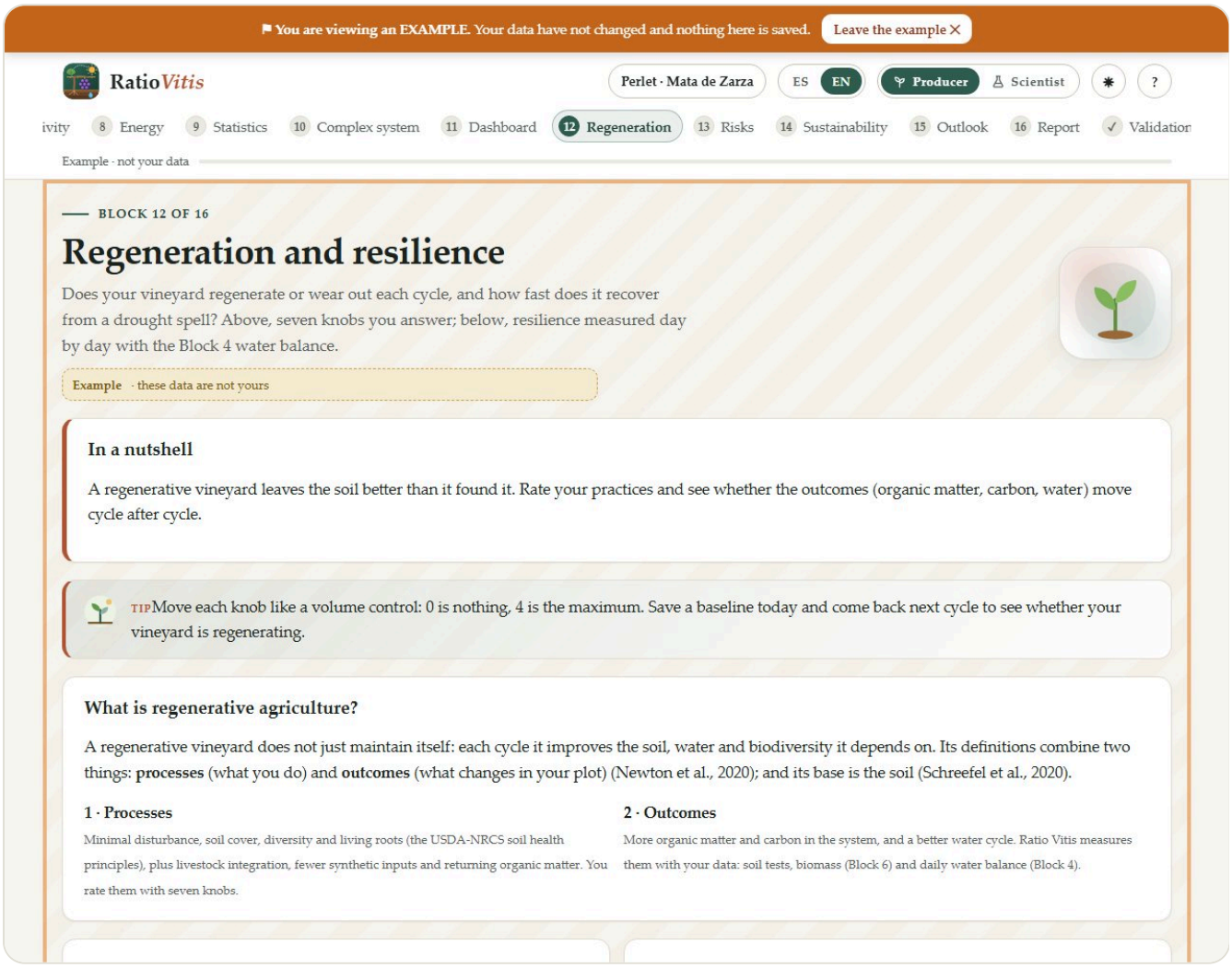


FIGURE 12.1 **Regeneration and resilience.** The knobs start unscored: nothing comes pre-filled.

BLOCK 13 · RISKS

13 Likelihood times impact

How likely and how harmful each problem would be.

For each risk (drought or late rains, heavy rain, cyclones, pests, fungi, price falls, labour shortage…) move likelihood and impact from 1 to 5 and say how much control you have. Beside each you see what your data say, such as Block 4 stress days. The matrix paints in red what to deal with first.

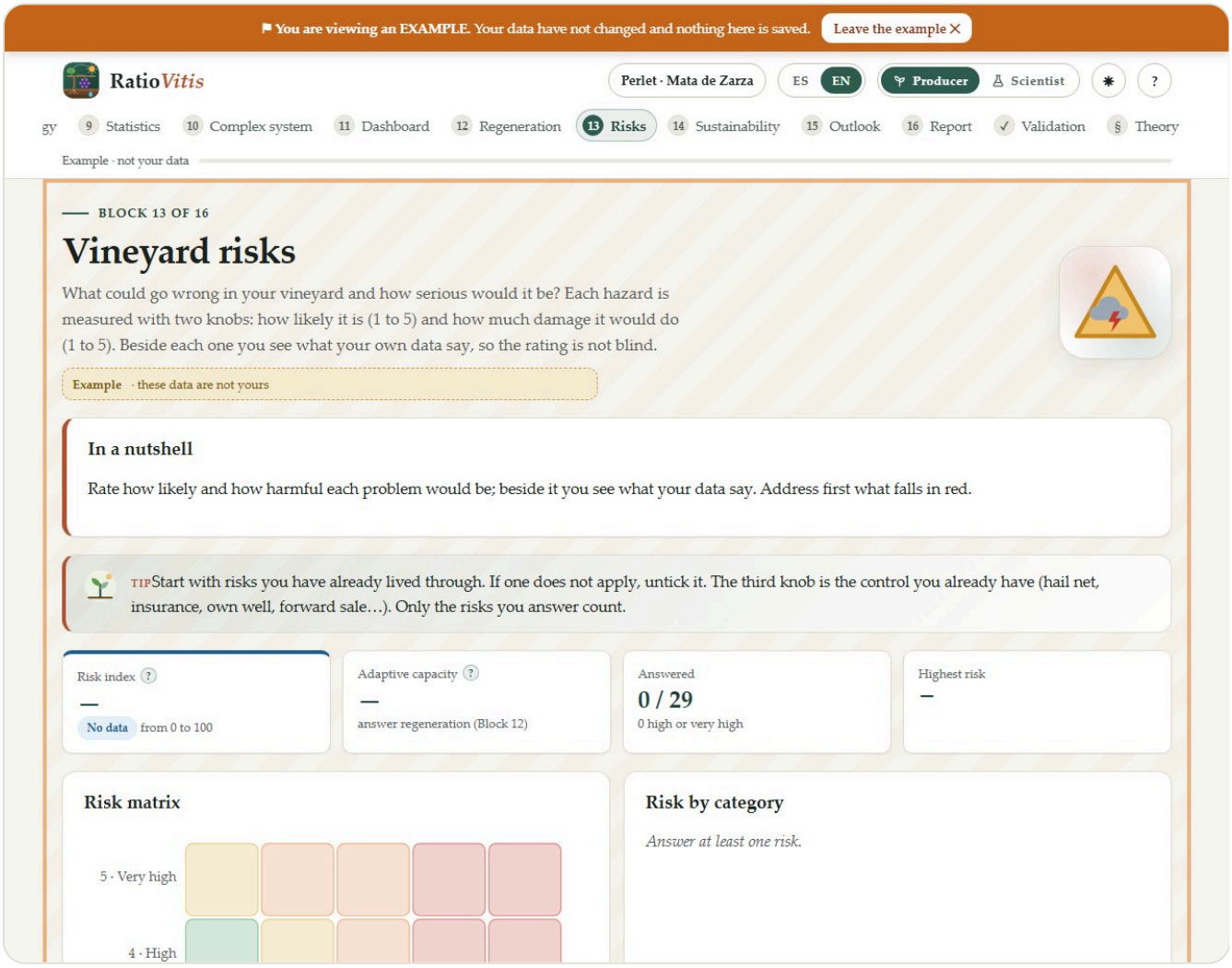


FIGURE 13.1 **Vineyard risks.** The matrix fills in as you score.

BLOCK 14 · SUSTAINABILITY

14 Which treatment is more sustainable?

MESMIS, Sarandón and the AMIBA.

Compare treatments in three dimensions —environmental, economic and social— with the thesis indicators (organic matter, carbon, water efficiency, yield, cost, benefit/cost, labour, carbon footprint…). Each indicator is normalised 0 to 1; a treatment is **sustainable** with a general index (ISG) of 0.5 or more and no dimension below 0.5 (Sarandón, 2002). The AMIBA plots all treatments' indicators on a radar.

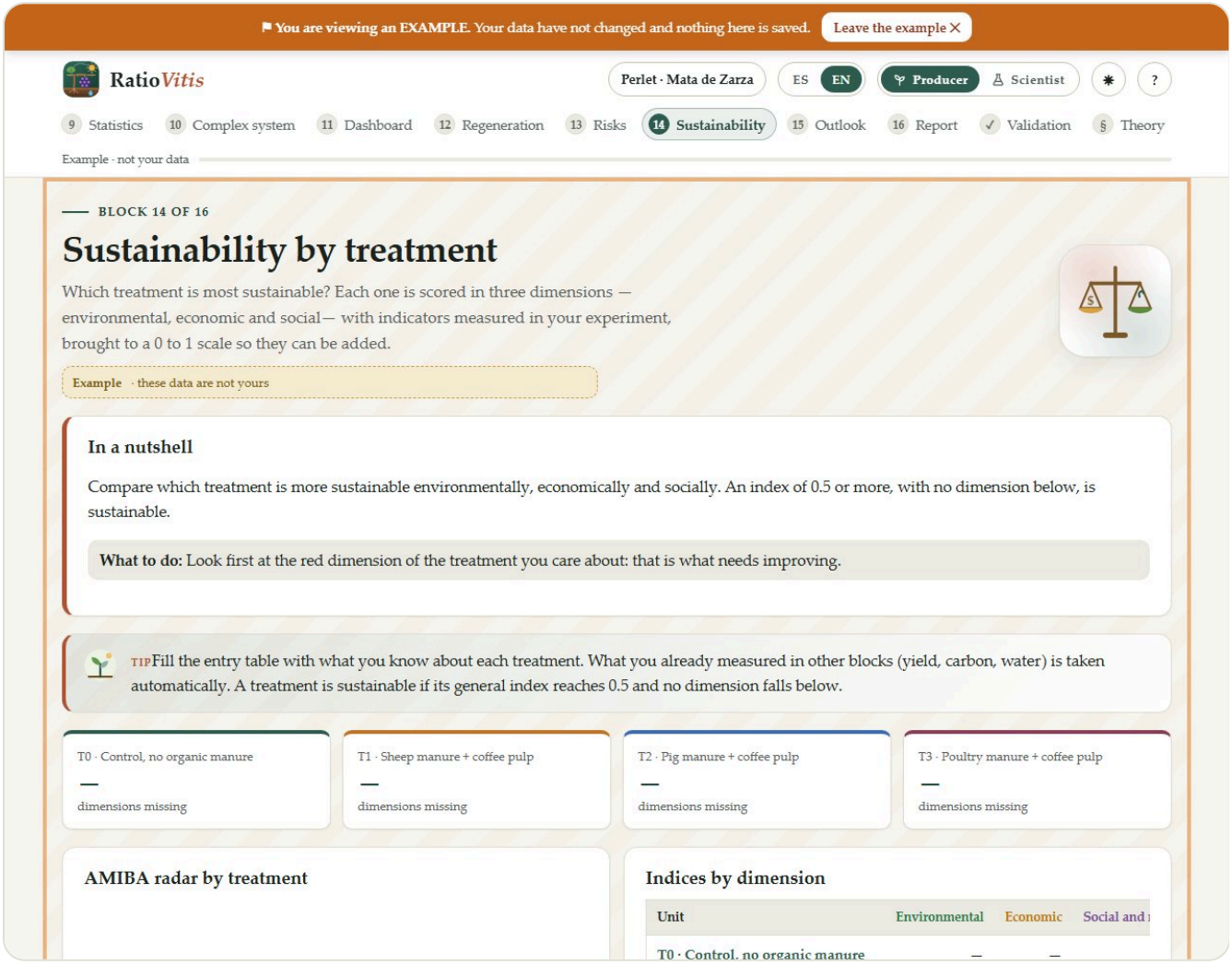


FIGURE 14.1 **Sustainability by treatment.** Computed once you enter the 14.1 indicators.

BLOCK 15 · OUTLOOK

15 At 1, 2 and 5 years

Which treatment best withstands drought and stays sustainable.

Three possible futures —desirable, trend and undesirable— at 1, 2 and 5 years. Enter how much each treatment's organic matter rises or falls per year: it is what most changes the water your soil holds. The app recomputes water retention (Saxton and Rawls, 2006), reruns the Block 4 balance with each scenario's climate and recomputes the sustainability index. That is how the thesis hypothesis is tested: do treatments that raise organic matter give more resilience?

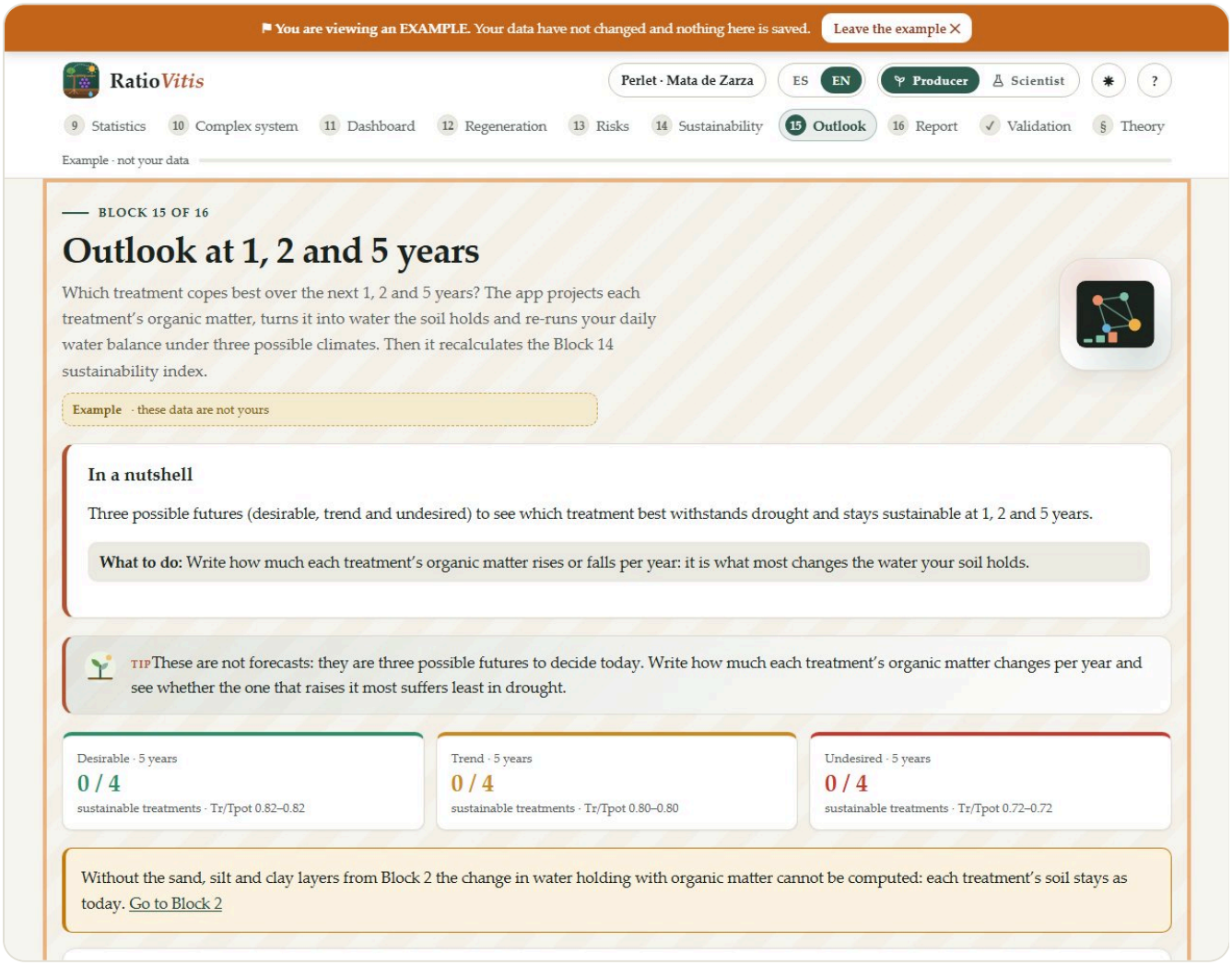


FIGURE 15.1 **Outlook.** It needs the jar test and each treatment's organic matter.

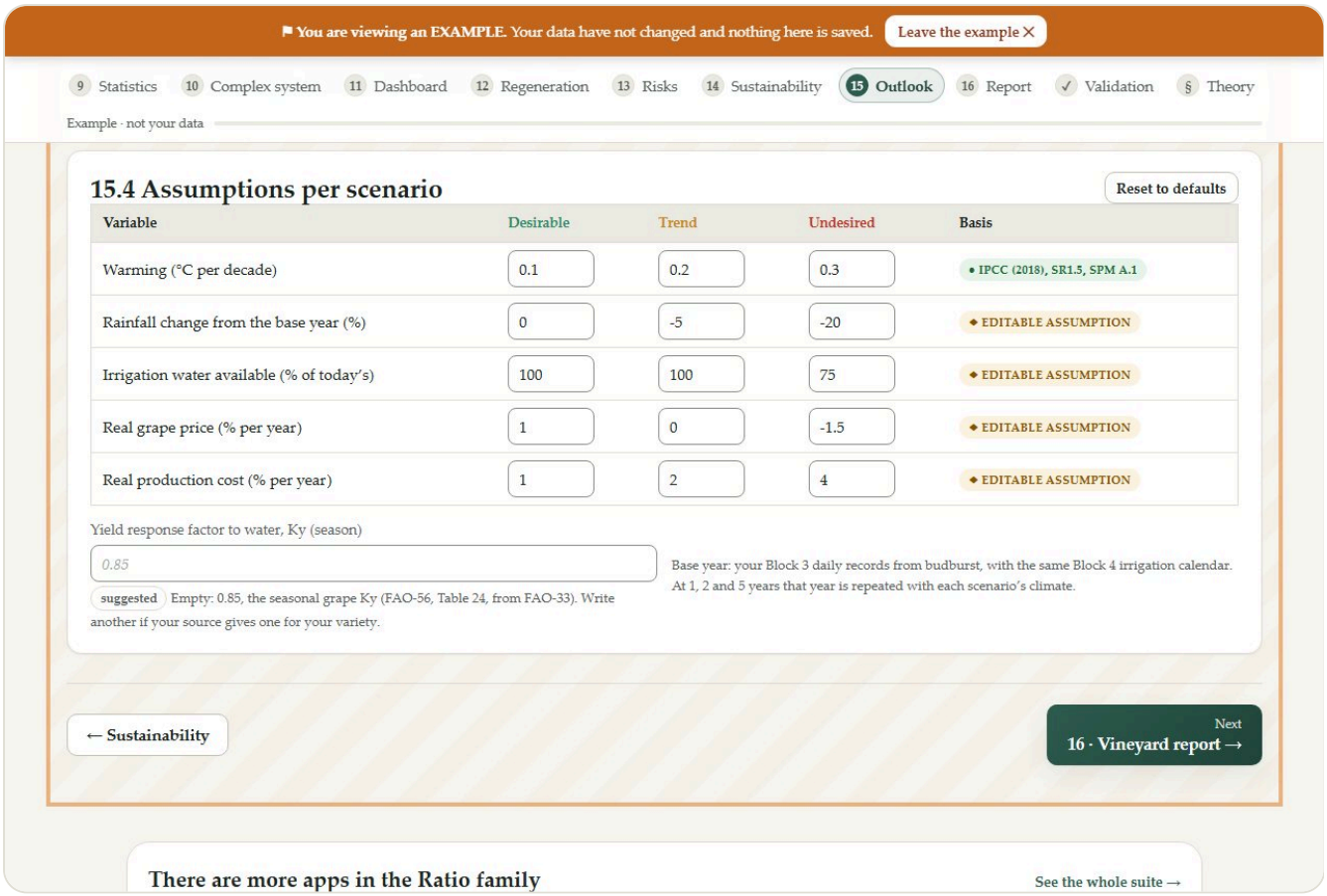


FIGURE 15.2 **15.4 Each scenario's assumptions.** Warming comes from IPCC (2018); the rest are flagged as editable.

> **!** **SCENARIOS, NOT FORECASTS**
>
> A scenario says what would happen if its assumptions hold. Change the assumptions and you will see how much the result depends on them.

BLOCK 16 · REPORT

16 Actions, PDF and data

What you take to the meeting.

Your vineyard in a document for the meeting, the adviser or the thesis committee: a cover with a summary written only from what was entered, an **action sheet** (each action says which block it comes from), Dashboard, water and phenology, sustainability and outlook, with the citation and QR code. Print or save as PDF with the button at the top; download the per-treatment, daily-balance and outlook CSVs, and save the project.

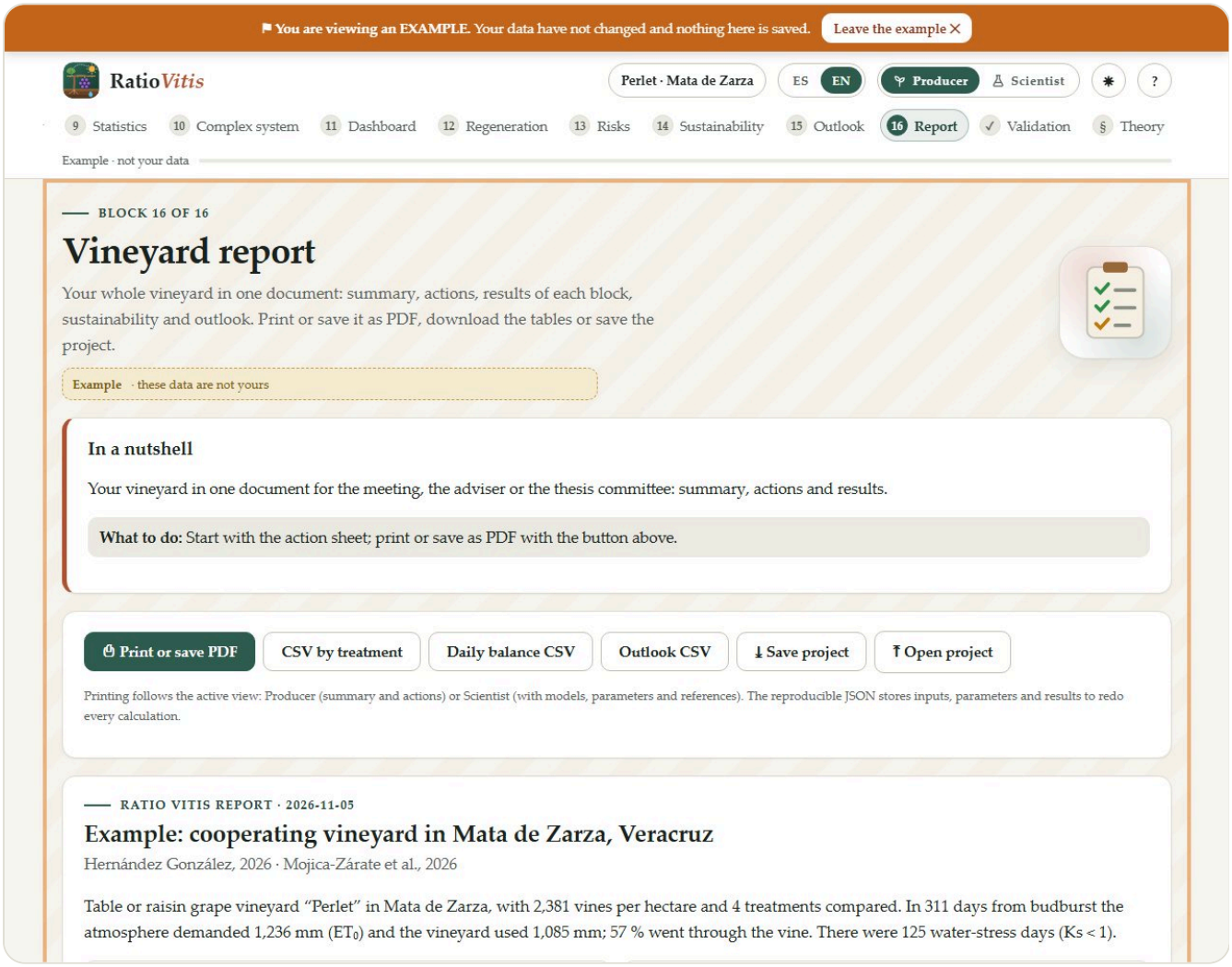


FIGURE 16.1 **Mata de Zarza report.** The action sheet starts with the 125 stress days.

✓ **SAVE THE PROJECT OFTEN**

The project file is your backup and what you share with your adviser. Save it after every field visit.

REFERENCE

A Field sheet

For notes in the vineyard.

Print this sheet and take it to the vineyard; one column per visit. What you write here is what you enter in blocks 4 to 7 and in the Block 9 sheet.

VARIABLE	VISIT 1	VISIT 2	VISIT 3	VISIT 4
Row / block				
Plot and treatment				
Vine (number)				
Age (years)				
Longest shoot (cm)				
Pruning weight (kg)				
NDVI				
Stem potential (MPa)				
Irrigation (L per vine)				
Flowering / veraison (date)				
Clusters				
Yield (kg)				
°Brix				

REFERENCE

B If something goes wrong

Common problems and their fix.

WHAT YOU SEE	WHY	WHAT TO DO
“no data” in a result	An entry is missing in an earlier block.	The notice says which; tap it to go there.
The balance does not appear	Climate, site, grape type, soil, stages or budburst missing.	Block 4 lists what is missing.
The climate CSV will not import	Date, Tmax or Tmin columns missing, or Tmax below Tmin.	Check the headers; the app says how many rows it dropped.
An index reads “coverage”	Its period has under 90 % of days with data.	Fill in that period’s records.
The data disappeared	Browser storage was cleared or you changed device.	Open your last project file.

REFERENCE

C Glossary

The words of the measured vineyard.

Depletion (Dr) *Agotamiento (Dr)*	How much water the soil lacks, in millimetres, to return to field capacity. It is the net irrigation depth the app proposes.
RAW *AFA*	Readily available water: the part of available water the vine takes up without stress (p × TAW).
TAW *ADT*	Total available water in the root zone, between field capacity and wilting point, in mm.
AMIBA *AMIBA*	MESMIS radar chart comparing the sustainability indicators of each treatment.
ANOVA *ANOVA*	Analysis of variance: says whether the differences between treatments are larger than chance would give.
Block *Bloque*	In the trial, a group of similar plots (in Mata de Zarza, each row) where every treatment appears once.
Border vine *Bordo (planta de)*	A vine at the end of the row that receives the treatment but is not measured, to avoid edge effects.
Bowen ratio *Bowen (relación de)*	Sensible over latent heat. Low: the vineyard spends energy evaporating water; high: it heats the air.
Field capacity *Capacidad de campo*	The water the soil holds after excess has drained, one or two days after heavy irrigation.
Stress coefficient (Ks) *Coeficiente de estrés (Ks)*	Goes from 1 (no stress) to 0. Drops when depletion exceeds RAW (FAO-56).
DOP *DOP*	Deviation from optimum percentage of a nutrient: negative lacking, positive in excess (Montañés et al., 1993).
ET0 *ET0*	Reference evapotranspiration: how much water the air demands in a day, in mm, by FAO Penman–Monteith.

ETc *ETc*	Water the vineyard uses: vine transpiration plus soil evaporation.
GST *GST*	Growing season temperature, 1 April to 31 October in the northern hemisphere.
Huglin index *Índice de Huglin*	April–September heat sum that weighs sunny hours more (Tonietto and Carbonneau, 2004).
Ravaz index *Índice de Ravaz*	Kilos of fruit over kilos of pruning; between 5 and 10 the vine is balanced (UC Davis).
ISG *ISG*	General sustainability index, 0 to 1: sustainable at 0.5 or more with no dimension below (Sarandón, 2002).
Kcb *Kcb*	Basal crop coefficient: what share of ET0 the vine transpires in each stage (FAO-56, Table 17).
Ky *Ky*	Yield response factor to water; for grapes, 0.85 over the season (FAO-56, Table 24).
Ratio Vitis reading *Lectura de Ratio Vitis*	An app assumption to complete a calculation; always marked ◆ and editable.
Example mode *Modo ejemplo*	The Mata de Zarza case loaded separately: nothing is saved and your project does not change.
Plot *Parcela*	The experimental unit: a run of vines with the same treatment within a block.
Stem water potential *Potencial hídrico de tallo*	How much tension the water in the vine is under at midday, measured with a pressure chamber, in MPa.
Jar test *Prueba del frasco*	Shaking soil with water in a jar and measuring the sand, silt and clay layers as they settle.
Permanent wilting point *Punto de marchitez permanente*	The water left in the soil when the vine can no longer extract it.
SPAC *RASPA*	Soil–plant–atmosphere continuum: the path of water from rain and irrigation to the air, through the vine.

Reanalysis *Reanálisis*	Climate reconstructed from models and observations (such as ERA5); useful where there is no station, but not a measurement in your vineyard.
Feedback loop *Retroalimentación*	A network path that returns to its start: it balances (negative) or reinforces (positive).
Tukey *Tukey*	Test that compares treatments in pairs; different letters mean a real difference.
Winkler degree-days *Winkler (grados día)*	Heat useful to the vine from 1 April to 31 October, above 10 °C; defines regions I to V.

REFERENCE

Ratio Vitis on one page

The 16 blocks in one sentence each.

1

Vineyard and trial Locate the vineyard, enter the spacing and draw the treatments.

2

SoilJar test, bulk density, rooting depth, pH and salt.

3

ClimateImport the daily CSV and read ET0, Winkler, Huglin and GST.

4

SPACStages, budburst and irrigations; read the balance and stress.

5

FertilityTissue analysis, DOP and nitrogen-matched dose.

6

BiomassPruning, leaf area, NDVI, shoot and dry matter.

7

ProductivityYield, °Brix, Ravaz and loss from water shortage.

8

EnergyRead latent heat, sensible heat and Bowen.

9

StatisticsImport the field sheet and read ANOVA and Tukey.

10

Complex system Walk through the scenes and feedback loops.

11

DashboardStart with the traffic light.

12

RegenerationScore the practices and save a baseline.

13

RisksScore each risk; deal with the red ones.

14

SustainabilityEnter per treatment and read the ISG and AMIBA.

15

OutlookAdjust organic matter and compare three futures.

16

ReportPrint, export CSV and JSON, save the project.

Scan to open Ratio Vitis:
hectortecumshe-ai.github.io/RatioVitis

REFERENCE

E How to cite and credits

Authors, licences and case sources.

How to cite the app

Mojica-Zárate, H. T., Hernández-González, J., Díaz-José, O., & Ramírez-García, A. G. (2026). *Ratio Vitis: The measured vineyard as a complex adaptive system: soil–plant–atmosphere water relations, fertility, biomass, productivity and energy* (Version 0.1.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.24048724

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Case sources

Mojica-Zárate, H. T., Barrera-Guzmán, L. Á., Alejo-Bello, S., Partida-Sedas, J. G., Rojas-Reyes, J. O., Díaz-José, O., Tinoco-Rueda, J. Á., & Torres-Palacios, C. (2026). *Guía práctica para el establecimiento de uva en Mata de Zarza, Veracruz*. Chapingo Autonomous University.

Hernández González, J. (2026). *Adaptabilidad del cultivo de uva (Vitis vinifera L.) en Mata de Zarza, Veracruz, como sistema complejo adaptativo* (master's thesis proposal). Chapingo Autonomous University.

Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society, 146*(730), 1999–2049. https://doi.org/10.1002/qj.3803

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Héctor Tecumshé Mojica-Zárate · Jorge Hernández-González · Oscar Díaz-José · Adán Guillermo Ramírez-García

Chapingo Autonomous University, Mexico. Corresponding author ORCID (Héctor Tecumshé Mojica-Zárate): 0000-0002-9067-3983.

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